

AI-Assisted Smart Knee Wearable with Autoregulated NMES

Specification v3.0 and final MVP guidelines

VERSION	3.0, final MVP edition (supersedes the Phase 2 specification)
DATE	9 October 2026
STATUS	Draft for team review. Items marked G or "Decision needed" must be settled before submission.
DEVICE STATUS	Research prototype. Not a medical device. No stimulation has been applied to a person.
PREPARED	From the team's Phase 2 specification and form answers, with AI assistance. The team owns every value and claim.
TEAM	_____

EVIDENCE STATUS USED THROUGHOUT

V Verified **I Implemented** **S Simulated** **D Planned** **G Gap**

V is reserved for a requirement verified against its stated criterion by a recorded test (Part B.5).

How to read this document

This is the final-MVP edition of the specification. It keeps the structure of the Phase 2 document and does three things to it.

- **Corrects it.** An audit against the datasheets we know, the extremum-seeking paper and the form found 23 points to fix, from the controller module to the use of EMG during stimulation (Appendix E).
- **Labels the evidence.** Every requirement carries one of five statuses so that built, tested, simulated and planned are never mixed (Appendix A).
- **Adds the guidelines to finish.** Part B defines the MVP, the tests with pass criteria, the fault-injection matrix, the safe demonstration, the claims we may make and the cut-line.

Part A is the specification: what the device is and does. **Part B** is how to build, test, demonstrate and defend it. **Appendices** hold the traceability matrix, the cost tracker, the parameter register, the self-tuning study, the change log and the sources.

Evidence status

Status	Meaning	Example
V Verified	Verified against its stated criterion by a recorded test.	Model B export matches the Python model on 300 of 300 cases.
I Implemented	Implemented, not yet verified. Early tests the team reports (EMG on a real muscle, NMES module on a dummy load, angle filter tuning) count here until their records are checked against the criteria in B.5.	Hardware safety chain, firmware state machine.
S Simulated	Simulated only.	Model A 93.5 %, Model B 97 %.
D Planned	Designed or planned, not built or not yet run.	Posture gate, plan integrity, fault-injection campaign.
G Gap	Gap or decision needed.	BLE link, cost figures.

RULE · No stimulation on a person

In this MVP, stimulation goes into a dummy load only. Stimulation on a person starts only after ethics approval, clinical supervision and a pass of the safety-chain tests (T-HW-01 to 11). Every statement in this document is written to stay true under that rule.

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Executive summary

The device is a sensor-instrumented knee orthosis that delivers neuromuscular electrical stimulation (NMES) to the quadriceps during seated isometric sessions, twice a day for six weeks from 48 h after total knee arthroplasty. An AI layer suggests three session parameters and a physiotherapist approves each one.

Where the project stands. The orthosis is printed and assembled, the firmware is complete, the hardware safety chain is implemented, and the app, dashboard and backend run locally. Of 56 requirements, 1 is verified, 26 are implemented and unverified, 7 are simulated, 16 are planned and 6 are gaps (Figure 9). The next phase is verification, not more features.

What changes in v3.0

- **Control is re-scoped to what the sensors can support.** EMG read while current flows is dominated by the evoked response, so the 100 ms loop supervises current, and the assist level is set once per contraction from EMG measured with the stimulator off. The controller is a PI-type assist rule between a clinician floor and an approved ceiling (VI).
- **The MVP ships Mode 0 and Mode 1.** Mode 0 is a fixed dose, always available. Mode 1 is the EMG assist, enabled only when its checks pass. Extremum-seeking tuning and force control are roadmap items: they need a force signal (VI.1, Appendix D).
- **Current limits get margins.** The ceiling is capped at 50 mA so that $1.10 \times \text{ceiling}$ stays below the module maximum of 57.6 mA, the 60 mA firmware cut and the 65 mA hardware trip (IV.5).
- **New safeguards close audit gaps:** posture and motion gate, ATTEMPT state, reference MVC recorded at the clinic, stale-data detection, usage limiter, plan expiry and integrity.
- **Part B turns the specification into an MVP plan:** 51 tests (32 must-pass), 23 fault-injection cases, a 7-minute demonstration with fallbacks, and a claims register.

Five decisions that block the rest

ID	Decision	Recommended
D-01	Which ESP32 module is flashed	Name it. The S2 has one core and no Bluetooth.
D-02	Ceiling cap	50 mA with the margin chain.
D-07	ON time and ramp definition	3 s up, 15 s hold, 2 s down, as built.
D-08	Fallback if BLE is late	USB-serial plan transfer between sessions.
D-06	Approval cadence	One approval a day, 36 h validity.

The full list of 12 decisions is in B.10. Open items the team must confirm are marked with a red “Decision needed” box where they arise.

PART A · SPECIFICATION

I. Introduction

I.1 Context and problem statement

Total knee arthroplasty (TKA) is one of the most frequent orthopaedic procedures, and recovery depends on structured rehabilitation. The main barrier in the first weeks is quadriceps activation failure: a neuromuscular inhibition that persists whatever the patient's willingness to exercise. In the early-recovery literature, quadriceps strength falls by about 62 % at one month, with voluntary activation down 17 % and muscle cross-section down 10 % (Mizner et al., 2005). This delays the recovery of range of motion (ROM), gait and independence.

Neuromuscular electrical stimulation (NMES) is a published countermeasure, but common units run in open loop: intensity and timing follow a preset programme, without feedback. That risks under-stimulation, which limits the benefit, or over-stimulation, which causes discomfort and lowers adherence. Wearables that monitor ROM and adherence exist, but they observe and cannot act.

Existing solution	Limitation for early TKA rehabilitation
Supervised physiotherapy	Limited session time; no measurement of activation at home
Continuous passive motion (CPM)	Passive: does not train the quadriceps
Conventional NMES units	Open loop: fixed intensity regardless of the patient's effort; no data
EMG biofeedback devices	Measure only; no stimulation, no knee angle
Remote-monitoring apps and wearables	Monitor ROM or adherence; no therapeutic action

I.2 Objective

The project unifies sensing and stimulation in one wearable: a sensor-instrumented knee orthosis that delivers NMES to the quadriceps in seated isometric sessions during the first 6 weeks after unilateral TKA, with the dose set by the patient's measured voluntary effort inside limits that a clinician approves. Before each day's sessions, an AI layer suggests three parameters from the patient's own data, and the physiotherapist validates each value.

MVP objective. For the SymbioMed Challenge, the objective is a working, safe-to-demonstrate prototype: the full chain from plan approval to a session on a *dummy load*, with the evidence to show what works and honest labels on what does not yet. It is not a claim of clinical benefit (B.1).

I.3 Scope and eligibility

- **Scope.** Passive printed orthosis, electronics, firmware, AI layer, mobile app, web dashboard and backend. NMES runs in two seated isometric sessions a day for 6 weeks. Walking stays part of standard rehabilitation and is not stimulated. Certification and clinical trials are outside this phase.
- **Users.** Adults after primary unilateral TKA (typically 55–85 years) at the clinic and at home; physiotherapists who validate each prescription and follow sessions remotely.
- **Eligibility.** Surgeon clearance for NMES from 48 h after surgery. Contraindications are listed in X.3: pacemaker or implanted stimulator, active deep vein thrombosis, damaged skin, wound or infection under the electrodes, impaired circulation, malignancy in the treated area, uncontrolled epilepsy, and inability to use the STOP button.
- **Not in scope for the MVP.** Stimulation on any person; clinical claims; hosted data storage; regulatory submission.

I.4 Where the MVP sits in the recovery timeline

The device targets the early and middle phases of rehabilitation, from 48 h after surgery to week 6, when voluntary quadriceps activation is at its lowest and NMES is used in the cited protocols. Figure 1 places the device against the usual phases; timings vary by surgeon and hospital.

time after surgery, not to scale →

	Pre-op prehab	Day 0–2 inpatient	Day 2 – week 2 early	Weeks 2–6 middle	Weeks 6–12 late	Week 12 + long term
PATIENT general clinical knowledge	Painful, stiff knee; often weak quadriceps	Pain on medication, swollen knee, needs help to transfer and walk	Moderate pain and swelling, walker; voluntary quadriceps activation at its lowest	Pain and swelling fall; walker to cane; bend reaches about 90–105°	Walks without a device; stairs get easier	Most daily tasks restored; some residual weakness
REHAB FOCUS from the cited protocols	Strength, mobility, learn the equipment	Therapy starts within 24 h: transfers, walking aid, quad sets	Bend about 15–90°; quad sets, straight leg raises, short walks	Bend about 10–105°; bike, seated extension, sit-to-stand	Progressive strengthening, closed-chain work, active bend toward 120°	Balance, movement quality, low-impact activity
NMES IN PROTOCOLS	Optional (home prehab pilots)	Not yet: surgeon clearance typically from 48 h	Started at 48 h in the reference trial, twice daily	Continued to week 6 in the reference trial	Weaned as quadriceps control improves	Normally discontinued
THIS DEVICE (v3.0)	Not used	Off	Mode 0 with a titrated ceiling; Mode 1 only if voluntary EMG is measurable	Mode 1 (MVP target). Mode 2 and tuning: roadmap	Off by default; any extension needs a clinician and new evidence	Off; monitoring-only use is future work

MVP scope: 48 h to week 6, seated isometric sessions, twice a day

Patient and rehab rows summarise general clinical knowledge and the protocols listed in Appendix G; timings vary by surgeon and hospital.

Figure 1. Recovery phases, rehabilitation focus, NMES use in the cited protocols, and what this device does in each. The MVP covers 48 h to week 6.

II. System overview and design philosophy

II.1 Design philosophy

The governing idea is unchanged from Phase 1: stimulation should respond to the muscle's measured behaviour rather than follow a fixed programme. Four principles shape what the device is allowed to do.

- The AI suggests; the clinician decides.** The AI never commands the stimulator. Every plan is approved, and firmware limits apply whatever the plan says.
- Every safety-critical function works in hardware as well as firmware.** The STOP button, USB lockout, over-current trip, watchdog and enable gate work without the processor.
- Control matches what the sensors can support.** Surface EMG read while current flows is dominated by the evoked response, so voluntary EMG is read only with the stimulator off (VI.1). This is a change from v2.
- Always fall back to the simple, safe mode.** A fixed dose (Mode 0) is available whenever a check fails (VI.1).

The system has four layers that run as a cycle: sensing, a decision layer (signal processing, an assist rule, an independent safety supervisor, and the AI that prepares each day), an actuation layer (a constant-current biphasic output), and a connectivity layer (BLE, app, dashboard, backend) that never interrupts the real-time loop.

II.2 Architecture

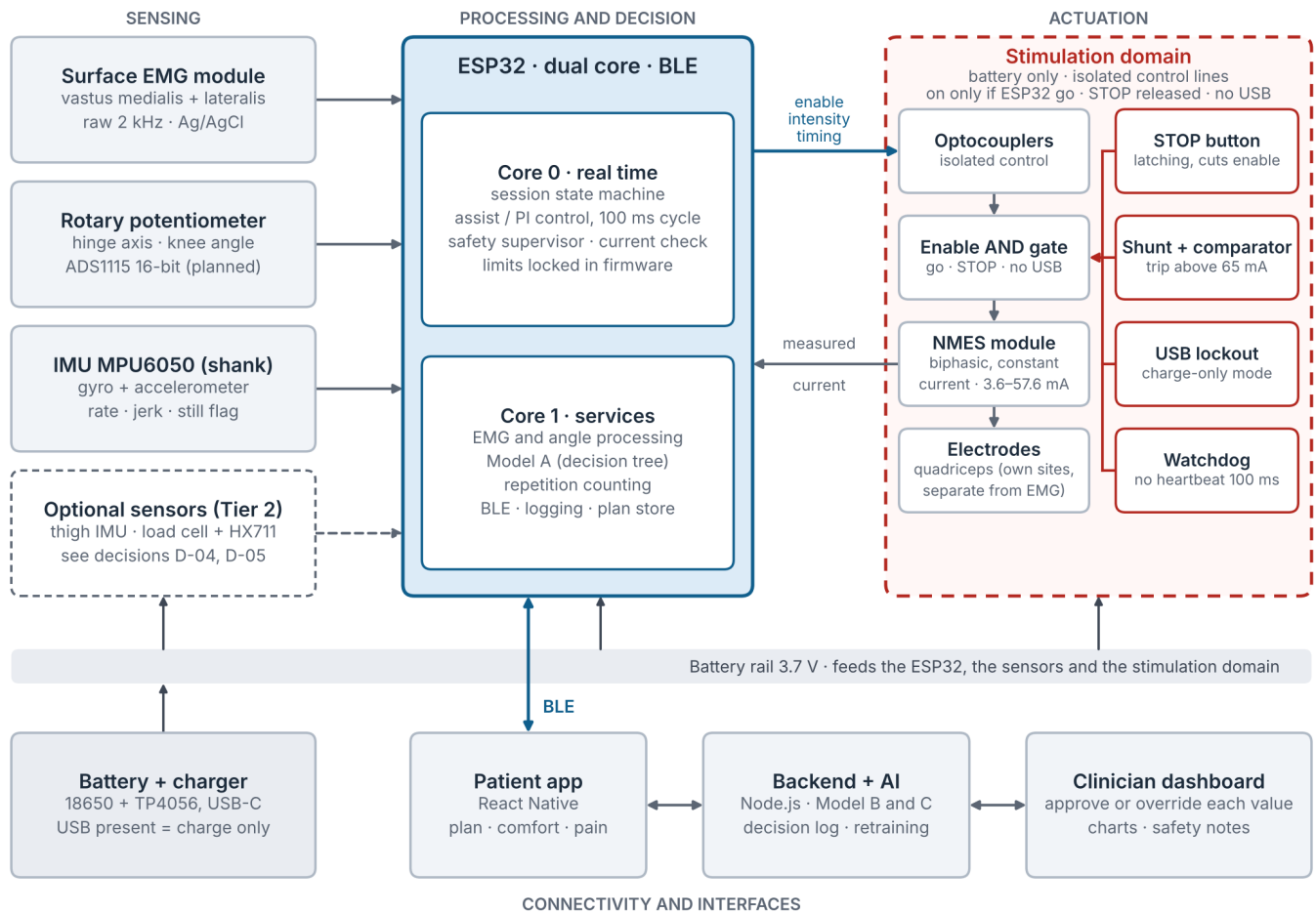


Figure 2. System architecture: sensing, decision, actuation, power and connectivity. The dashed boundary is the stimulation domain: battery-powered only, with isolated control lines, enabled only by the controller, a released STOP button and no USB connection.

The controller runs the safety-critical loop on a single processor core, separate from communication. The AI that prepares each session runs on the backend, outside that loop, and never commands the stimulator directly. The controller must have two cores and Bluetooth: that means a classic ESP32 or an ESP32-S3, not the ESP32-S2 named in v2, which has one core and no Bluetooth (D-01).

DECISION NEEDED · D-01 · Controller module

Write the exact module that is flashed (classic ESP32 or ESP32-S3) into the specification and the bill of materials. If the board is really an S2, the two-core split and BLE cannot work as written.

III. Functional specifications

III.1 Main functions

Each function has a measurable target and a status. The status is the evidence available today (Appendix A has the full traceability matrix, with 56 requirements).

ID	Function and criterion	Status
MF1	Measure quadriceps activation: RMS envelope of the surface EMG in % of the reference MVC, accepted only when the gate is open	I Implemented
MF2	Measure knee angle over 0–120° at 100 Hz, within ±3° of a goniometer, and count repetitions	I Implemented
MF3	Assess fatigue and movement quality: median-frequency drop in valid windows, IMU jerk, Model A state per contraction (≥ 90 % correct)	S Simulated

ID	Function and criterion	Status
MF4	Deliver controlled NMES: plateau current set once per contraction between floor and ceiling, supervised every 100 ms, delivered current within $\pm 10\%$ of command	I Implemented
MF5	Suggest the next prescription (ceiling, OFF time, contractions) with reasons; ceiling within 5 mA in $\geq 90\%$ of cases	S Simulated
MF6	Let the clinician validate every parameter; 100 % of sessions use an approved plan	I Implemented
MF7	Record and share sessions: summary to the app over BLE, comfort 0–3 and pain 0–10; no session lost after sync	G Gap
MF8	New: gate every measurement and every contraction start (knee $60^\circ \pm 10^\circ$, shank still, stimulator off, EMG valid)	D Planned
MF9	New: demonstration mode (stimulation into a dummy load only, volunteer EMG drives the command)	D Planned
MF10	New: enforce usage limits (at most 2 sessions a day, minimum gap, plan expiry)	D Planned

III.2 Constraints

ID	Constraint and target	Status
CF1	NMES parameters locked in firmware: 50 Hz, 150–400 μ s, ramps 3 s up and 2 s down, hold 15 s, OFF 30–90 s, ceiling ≤ 50 mA	I Implemented
CF2	Electrical safety: battery-only stimulation, isolated control lines, current limits, lead-off detection	I Implemented
CF3	Emergency stop: hardware cut in less than 10 ms, latched until restart	I Implemented
CF4	Human oversight: every plan approved by a clinician; the last approved plan is kept within its expiry	I Implemented
CF5	Comfort and fit for adults 55–85; patella and electrode sites visible; STOP reachable	I Implemented
CF6	Autonomy: two sessions of up to 40 min per charge	D Planned
CF7	Cost at or below 575 TND per prototype	G Gap
CF8	Designed with reference to IEC 60601-1, IEC 60601-2-10 and ISO 14971; dummy-load tests before any human use	D Planned
CF9	New: data protection (role-based access, hashed passwords, expiring tokens, no personal data in logs)	G Gap
CF10	New: a first-time user can fit the brace, start a session and find STOP without help	D Planned

IV. System architecture

IV.1 Sensing layer

- **Surface EMG.** The signal is weak (0.5–10 mV), with useful content between 20 and 450 Hz, and its spectrum shifts lower as the muscle fatigues. A ready-made EMG module, tested on a real muscle, replaces the custom circuit of Phase 1. Electrodes sit over vastus medialis and vastus lateralis, separate from the stimulation electrodes. During stimulation the recording is dominated by the evoked response and artefact; the device therefore uses voluntary EMG only with the stimulator off (V.3).
- **Rotary potentiometer.** Single turn, low torque, 10 k Ω , coaxial with the hinge. A $\pm 1\%$ linearity over about 270° of travel is about $\pm 2.7^\circ$, which leaves no budget for the $\pm 3^\circ$ target, so v3.0 adds a 5-point calibration against a goniometer and moves the reading to an external 16-bit ADS1115 converter (IV.4).
- **IMU (MPU6050) on the shank shell.** Gives angular velocity, movement smoothness and a shank-still flag, and cross-checks a potentiometer that could slip. It has no magnetometer, so it cannot give hip flexion or trunk lean (D-05).

IV.2 Processing and decision layer

The controller turns raw signals into voluntary activation, fatigue index, knee angle, repetitions and movement quality (V). Three decision blocks act on them.

- **Assist rule (real time).** Once per contraction, sets the plateau current between a clinician floor and the approved ceiling, using voluntary EMG measured with the stimulator off (VI.2). The 100 ms loop supervises current and safety conditions.
- **AI personalisation (between sessions).** Model A labels each contraction on the controller; Model B on the backend suggests the next prescription, which the clinician validates (VIII).
- **Safety supervisor.** An independent rule-based module that enforces hard limits and stops stimulation on faults, whatever the assist rule or the AI computes (X).

IV.3 Actuation, structure and fit

- **Actuation.** A commercial constant-current biphasic NMES module (3.6–57.6 mA). The controller sets enable, intensity, pulse width, ramp and timing over galvanically isolated lines (optocouplers), and the output current is measured and fed back (VI.5).
- **Structure.** FDM-printed thigh and shank shells in PETG with TPU padding and a printed hinge. The thigh cuff holds the electronics; an anterior channel keeps the patella and electrode sites visible; the STOP button is within one-handed reach. Adjustable for adults 55–85 (CF5).

IV.4 Connectivity and component summary

BLE connects the brace to the mobile app. The app uploads each session summary to a Node.js backend that runs Model B, stores the clinician's decisions and serves the approved plan the brace reads at the next sync. A USB-serial path with the same plan format and integrity checks is the fallback if BLE is late (D-08); it works between sessions only, because USB present means no stimulation.

Role	Component	Key specification	Note
Controller	ESP32 (classic or S3), D-01	Two cores, BLE	S2 does not fit
Muscle activation	Commercial surface-EMG module, Ag/AgCl electrodes	Raw EMG, 20–500 Hz band	Tested on a real muscle
Knee angle	Rotary potentiometer 10 kΩ + ADS1115	5-point calibration	ESP32 ADC is non-linear
Motion	MPU6050 (GY-521) on the shank	3-axis gyro + accelerometer, I2C	Gives still flag and jerk
Stimulation	Commercial NMES/EMS module	Biphasic, constant current, 3.6–57.6 mA	Tested on a dummy load
Safety	Shunt and comparator, optocouplers, latching STOP, enable AND gate, watchdog	Cuts output without firmware	Section X
Power	Protected 18650 + TP4056 USB-C	3.7 V nominal	USB present: charge-only
Structure	Printed PETG/TPU orthosis, textile straps	Fast, cheap fit iteration	

IV.5 Power, autonomy and limits

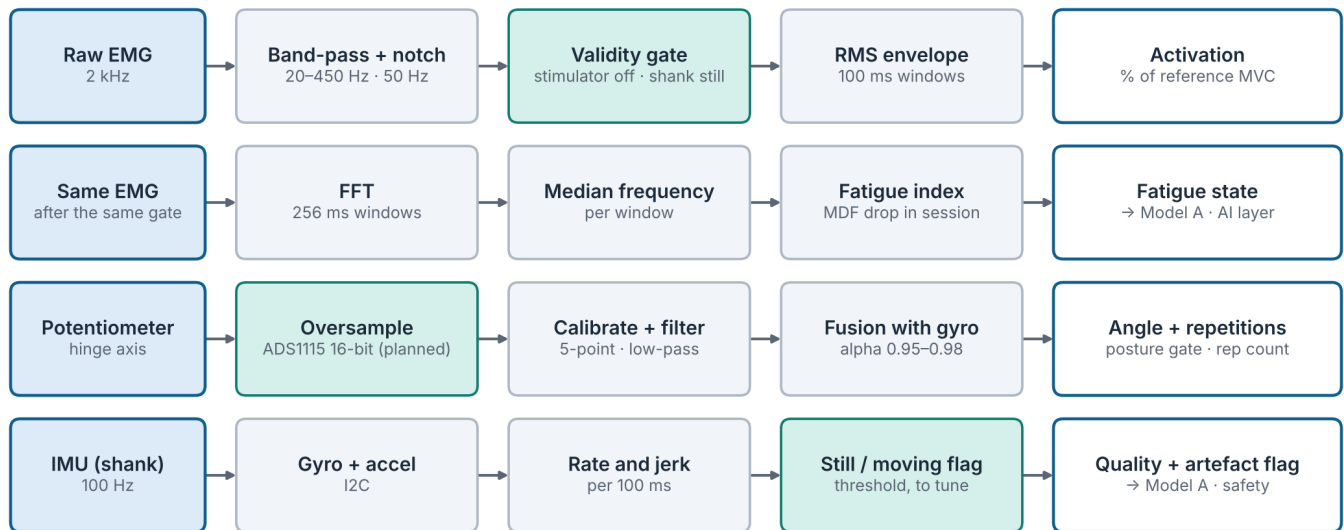
CF6 asks for two sessions of up to 40 min per charge. The longest plan the firmware accepts is 20 contractions at 3 s ramp up, 15 s hold, 2 s ramp down and 90 s OFF: $20 \times (3 + 20 + 90) \text{ s} = 37.7 \text{ min}$. A 40 min allowance therefore covers the longest plan with 2 min to spare, but the figure must be confirmed by a runtime test (T-PWR-01); today it is a design value **D Planned**.

The stimulation current limits are chained so that each protection sits above the one before it with room to spare, shown in VI.5. Parameter limits, thresholds and basis for every number are in Appendix C.

V. Signal processing

V.1 Pipelines

Four pipelines turn the raw signals into the features used by the decision layer.



added or changed in v3.0

Figure 3. Signal processing pipelines. The EMG amplitude branch feeds the assist rule only from the ATTEMPT window, when the stimulator is off.

- **Potentiometer.** Read directly at the hinge through the ADS1115, oversampled, calibrated at 5 points and low-pass filtered. Repetitions are counted by threshold crossing.
- **IMU and fusion.** A complementary filter combines the potentiometer angle (absolute, noisy) with the gyroscope rate (precise, drifts):

$$\theta_k = \alpha(\theta_{k-1} + \omega \Delta t) + (1 - \alpha) \theta_{\text{pot}}$$

with α between 0.95 and 0.98, tuned against a goniometer. Angular velocity and jerk give movement smoothness.

- **EMG amplitude branch.** Band-pass 20–450 Hz, notch 50 Hz, rectification and RMS over 100 ms windows with a 50 ms hop. Spikes that coincide with IMU acceleration transients are rejected as motion artefacts. Normalised to the reference MVC (V.3). This is the input of the assist rule, read only with the stimulator off.
- **EMG frequency branch.** FFT over 256 ms windows and median frequency; its drop across the session is the fatigue index, computed in valid windows only. Evoked EMG changes with fatigue differently from voluntary EMG, so windows during stimulation are not mixed in.

V.2 Rates and timing

Signal	Rate	Justification
Raw EMG	2000 Hz	Band up to 450 Hz with margin; keeps the spectrum for fatigue analysis
EMG envelope	20 Hz	100 ms windows, 50 ms hop (v2 said 10 Hz; 50 % overlap gives a value every 50 ms)
Potentiometer, IMU, fusion	100 Hz	Knee motion is below 5 Hz
Control and supervision loop	10 Hz (100 ms ± 5 ms)	Supervises current, gates and faults
Stimulation current	Every pulse	Checks each pulse delivered
Stale-data timeout	500 ms	Design value: EMG, angle or IMU frozen or missing longer than this stops the session

V.3 Gates, validity and reference MVC

A measurement is used, and a contraction is started, only when its gate is open. The gate is evaluated in POSTURE_CHECK before every contraction and again in ATTEMPT.

Check	Condition	If it fails
Posture	Knee $60^\circ \pm 10^\circ$, seated instruction acknowledged	No start; message to the patient
Still	Shank angular rate below $10^\circ/\text{s}$ for 1 s (to tune in T-SEN-05)	Attempt rejected; repeated up to 3 times
Stimulator off	No current for the guard time of 1.0 s (design value; T-SEN-04b decides)	EMG not valid
EMG valid	Noise floor inside its calibrated window; no saturation	Hold last value, then Mode 0 at the floor
Sensors fresh	No sensor older than 500 ms	SAFE_STOP

Reference MVC. The reference maximal voluntary contraction is recorded at the clinic as the best of three efforts and stored in the plan. It is updated upward only, and only when the clinician records a higher value (D-03). Home sessions use the stored value. A per-session MVC would be unreliable when pain limits the effort, and unsafe as a reference for a dose.

Lead-off. A lead is flagged off when the noise floor leaves its calibrated window for more than 1 s. Flat EMG is not used as the criterion, because early after surgery a healthy electrode can show almost no voluntary activity.

VI. Stimulation control

VI.1 Control modes and the two loops

v2 described a PID updated every 100 ms from EMG during stimulation. That is not supported by the sensors: while current flows, the recorded EMG is dominated by the evoked response, which follows the current rather than the patient's effort. v3.0 therefore separates two loops (Figure 5) and defines three modes (Figure 4).

Session start checks (SELF_TEST, posture, MVC reference, artefact-test result, sensors present) select the highest mode whose checks all pass.

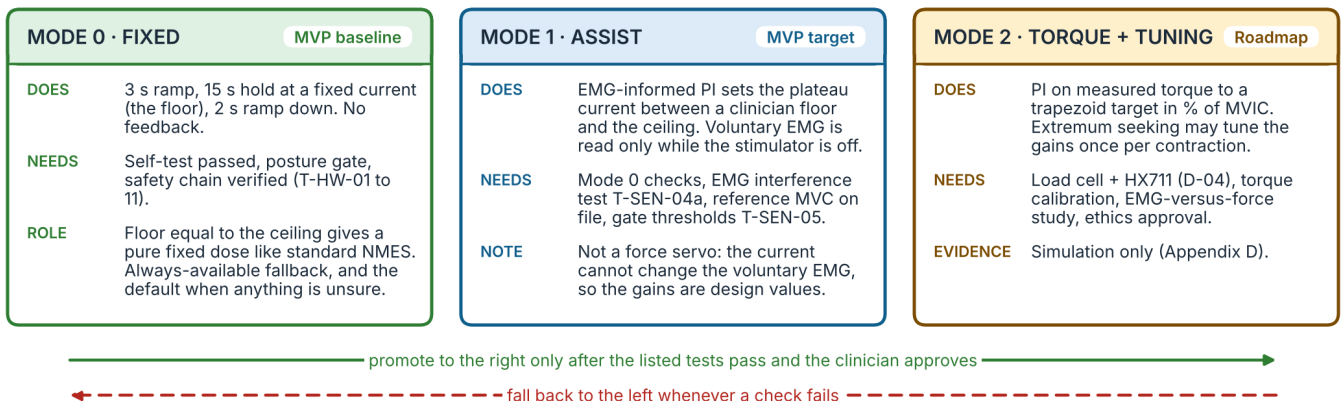


Figure 4. The three control modes. The session start checks select the highest mode whose checks all pass; any failing check falls back to the left.

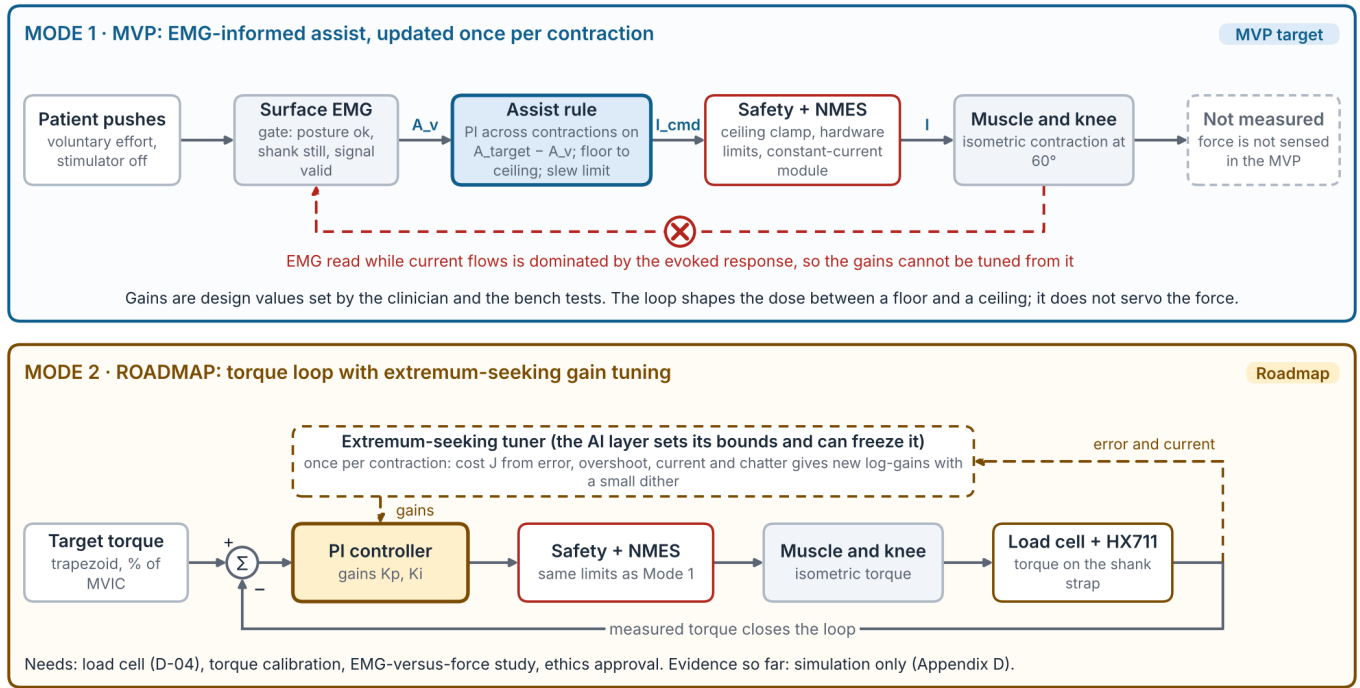


Figure 5. The two loops. A slow loop sets the plateau once per contraction from voluntary EMG; a fast loop supervises current every 100 ms.

WHY · The controller does not tune itself from EMG

Voluntary EMG read with the stimulator off does not depend on the stimulation current, so no method that perturbs the current can learn from it. Extremum seeking (Appendix D) needs a signal the current changes: force or torque, or the evoked EMG. That is why Mode 2 needs a load cell and is on the roadmap, not in the MVP.

Mode 0 (MVP baseline). A fixed dose at the floor current: 3 s ramp, 15 s hold, 2 s ramp down. With floor equal to ceiling it is a pure fixed dose like standard NMES. It is always available and is the default whenever anything is unsure.

Mode 1 (MVP target). The assist rule of VI.2 sets the plateau between floor and ceiling once per contraction. It is enabled only if Mode 0 checks pass, the EMG interference test (T-SEN-04a) has passed, a reference MVC is on file and the gate thresholds are set (T-SEN-05). It is not a force servo.

Mode 2 (roadmap). A PI on measured torque to a trapezoid target in % of MVIC, with extremum seeking tuning the gains. Needs a load cell and HX711 (D-04), torque calibration, an EMG-versus-force study and ethics approval.

VI.2 Assist rule (Mode 1)

Once per contraction k , during the ATTEMPT window with the stimulator off, the firmware computes the voluntary activation A_v as the highest 300 ms mean of the EMG envelope, as a percentage of the reference MVC, then updates a small PI-type state:

$$A_v(k) = \frac{100}{A_{\text{ref}}} \max_{t \in \text{ATTEMPT}} \overline{\text{env}}_{300\text{ms}}(t)$$

$$e(k) = \text{clip}\left(\frac{A_{\text{target}} - A_v(k)}{A_{\text{target}}}, -1, 1\right)$$

$$u_I(k) = \text{clip}(u_I(k-1) + K_I e(k), -u_{\text{max}}, u_{\text{max}})$$

$$x(k) = \text{clip}(x_0 + K_P e(k) + u_I(k), 0, 1)$$

$$I_{\text{target}}(k) = I_{\text{floor}} + x(k) (I_{\text{ceil}} - I_{\text{floor}})$$

$$I_{\text{cmd}}(k) = I_{\text{cmd}}(k-1) + \text{clip}(I_{\text{target}}(k) - I_{\text{cmd}}(k-1), -\Delta I_{\text{max}}, \Delta I_{\text{max}})$$

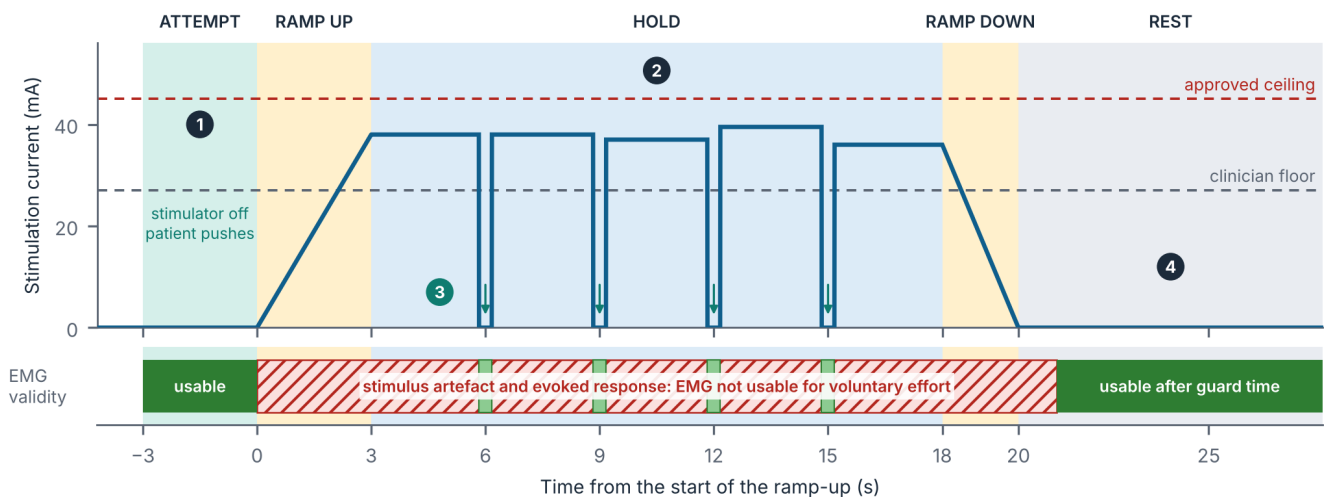
The commanded plateau changes by at most $\Delta I_{\text{max}} = 3$ mA between contractions. The first contraction of a session uses the floor.

Low voluntary activation (A_v below target) gives a positive error and pushes the plateau toward the ceiling; high activation lets it fall toward the floor. Because the update is once per contraction, a derivative term has no use and is dropped (v2 had one). Anti-windup is the clip on u_l .

Symbol	Default	Meaning
A_{target}	30 % of reference MVC	Clinician-set target for voluntary activation
K_p, K_i	0.5 and 0.1 per contraction	Design starting values, tuned on the bench
u_{max}, x_0	0.5 and 0.5	Integral limit and starting position within the window
I_{floor}	$0.6 \times I_{\text{ceil}}$ (0.4–1.0)	1.0 gives a fixed dose
ΔI_{max}	3 mA per contraction	Rate limit on the commanded plateau
I_{ceil}	10–50 mA	Approved by the clinician (D-02)

Special cases. Fatigue (Model A) holds the current; guarding stops stimulation and notifies the clinician. If the attempt window is invalid, the last A_v is held; after repeated invalid attempts the session runs in Mode 0 at the floor. All values are design values, not tuned or tested **D Planned**; the unit tests are T-FW-07 and the closed-loop test is T-SYS-02.

VI.3 One contraction



- Before the contraction (stimulator off): the patient pushes, EMG gives the voluntary activation A_v , and the controller sets this contraction's plateau current between the floor and the ceiling.
- During the contraction, every 100 ms: slew-limit and clamp the command, check the measured current, hold the last valid A_v . No new EMG data is used while current flows.
- Optional and off in the MVP (it needs test T-SEN-04b on a person): a short listening gap (about 120 ms, drawn wider here) refreshes A_v . The total gap time must stay below 5 % of the ON time.
- Rest: baseline-noise and lead-off checks, median-frequency fatigue index, Model A label, log the contraction, wait the approved OFF time.

Figure 6. One contraction: 3 s ATTEMPT with the stimulator off, then 3 s ramp up, 15 s hold, 2 s ramp down, then the OFF period. The current is supervised throughout.

Current flows for 20 s per contraction (3 + 15 + 2). With the 3 s ATTEMPT before it, a contraction takes 23 s plus the OFF time. The duty ratio, ON to OFF, is 1:1.5 at an OFF of 30 s, 1:2.25 at 45 s, 1:3 at 60 s and 1:4.5 at 90 s; the cited protocol uses a duty near 1:3 **D Planned** (D-07).

VI.4 NMES parameters and limits

Seven parameters are fixed at the values of a published post-TKA protocol; three follow recovery and are suggested by the AI and validated by the clinician. The firmware refuses any value outside these ranges.

Parameter	Value in this device	Clinical reference	Who sets it
Frequency	50 Hz	50 Hz at home, 50–75 Hz in clinic	Fixed
Pulse width	250 μ s default, range 150–400 μ s	250–400 μ s	Clinician, once at the first visit
Waveform	Biphasic, symmetric, constant current	Biphasic, current-controlled	Fixed
Ramp up, hold, ramp down	3 s, 15 s, 2 s	Ramps 2–3 s; ON 10–15 s	Fixed
OFF time	30, 45, 60 or 90 s	45–50 s (duty about 1:3)	AI suggests, clinician validates
Contractions per session	10, 15 or 20	15	AI suggests, clinician validates
Intensity ceiling	10–50 mA; plateau set between floor and ceiling	Maximum tolerated; at least 30 % of MVIC	AI suggests, clinician validates or overrides
Schedule	2 sessions a day, 6 weeks, from 48 h after surgery	Same	Fixed; usage limiter (VII.3)
Position	Seated, hip about 85°, knee 60°, isometric	Same	Fixed; posture gate (V.3)

References are those the team cited: the University of Delaware TKA protocol (2021) and Stevens-Lapsley et al. (2012). The protocol values here are as stated by the team and were not re-read (Appendix G).

VI.5 Output chain and current margins

The chain is: controller command → isolated control lines (optocouplers) → enable AND gate (battery only, STOP released, controller go, watchdog alive) → NMES module → electrodes, with a shunt measuring every pulse and a hardware comparator that can cut the enable line independently of the firmware. The control lines are galvanically isolated from the stimulation domain.

The ceiling cap (D-02) is chosen so the protections are in order, with margin below each:

$$1.10 I_{\text{ceil}} \leq 55 \text{ mA} < 57.6 \text{ mA} < 60 \text{ mA} < 65 \text{ mA}$$

Approved ceiling, tracking tolerance, module maximum, firmware cut and hardware trip, in mA. The cap is 50 mA, so $1.10 \times 50 = 55 \text{ mA}$.

A ceiling of 60 mA, as in v2, sits at the firmware cut and above the module's 57.6 mA maximum, so it could never be delivered and had no margin.

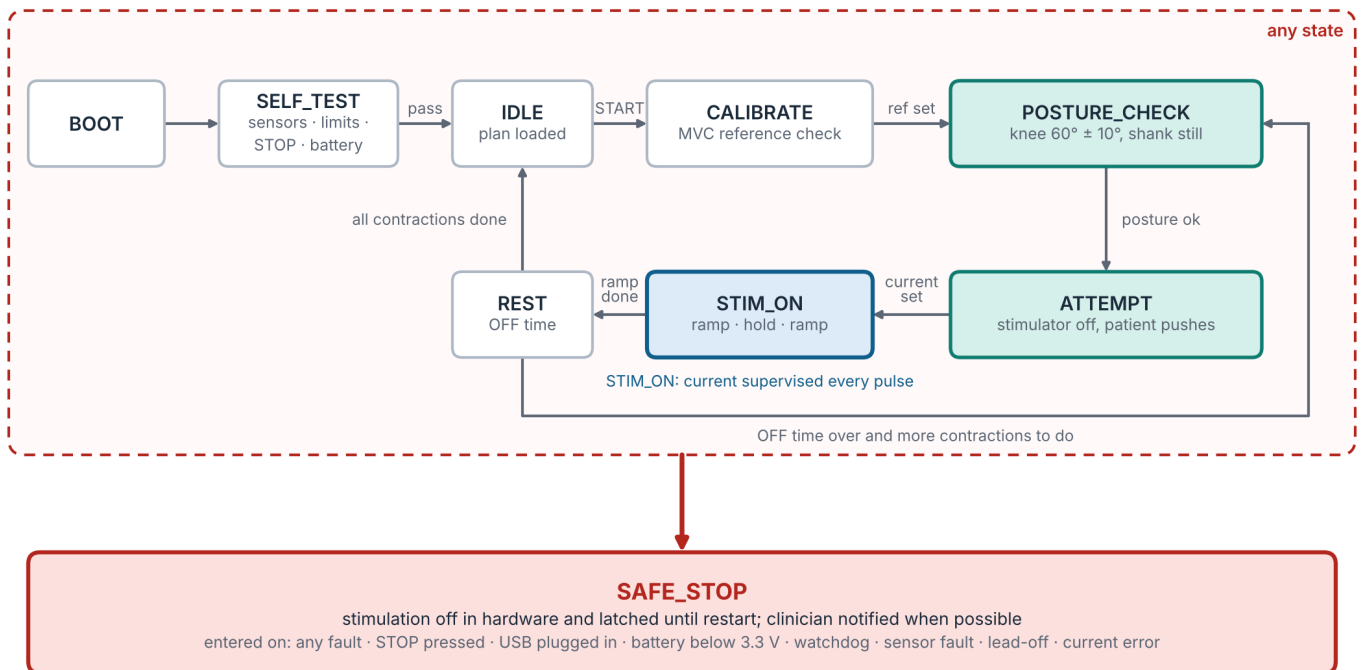
DECISION NEEDED · D-02 · Ceiling cap

Recommended: 50 mA with the chain above. Alternative: keep 60 mA and raise the firmware cut and the hardware trip, which means re-validating the trip. The dummy-load test range follows the choice (10–50 mA).

VII. Session operation

VII.1 Firmware and state machine

The firmware runs on the Arduino-ESP32 core as two tasks on separate cores, so a radio stall cannot delay a stimulation pulse. Core 0 runs the session state machine, the 100 ms supervision loop, current supervision and safety checks. Core 1 runs signal processing, repetition counting, Model A, BLE and logging. This needs a dual-core controller (D-01). A watchdog heartbeat from core 0 feeds the hardware watchdog.



 state added in v3.0

Figure 7. Firmware state machine (real-time core). POSTURE_CHECK and ATTEMPT are new in v3.0. Any fault, the STOP button, USB present, a battery below 3.3 V or a watchdog timeout forces SAFE_STOP.

A session loops POSTURE_CHECK → ATTEMPT → STIM_ON → REST until the approved number of contractions is done, then returns to IDLE. In STIM_ON, a 100 ms loop does four things: reads the measured current, compares it with the command, checks the gates and rate limits, and feeds the watchdog. It does not read EMG for control, because the EMG is not valid while current flows. SAFE_STOP cuts stimulation in hardware until the brace is restarted; restart never resumes a session. Power loss mid-contraction gives zero output and a restart in BOOT then SELF_TEST (T-HW-10).

VII.2 Usage scenario

First session at the clinic (day 2). The physiotherapist checks eligibility and the contraindication checklist, places the EMG and stimulation electrodes and fits the brace with the hinge on the knee joint. With the patient seated (hip about 85°, knee 60°), they set the pulse width (250 μ s, up to 400 μ s if no visible contraction), record the reference MVC as the best of three efforts, and set a first plan, for example a 20 mA ceiling, 45 s rest, 10 contractions, with the floor at 0.6 of the ceiling. After SELF_TEST, a short supervised session finds the maximum tolerated intensity, and the patient learns the STOP button and how to put on the brace.

Home sessions (twice a day, up to about 38 min).

1. The clinician approves the day's plan once (D-06). It is valid for 36 h by default. Until a new plan is approved, the brace uses the last approved plan within its expiry, and refuses to start after it.
2. The patient sits in the same position, puts on the electrodes and brace, and unplugs the charger. The app shows the approved plan.

3. SELF_TEST runs. For each contraction: POSTURE_CHECK, then the ATTEMPT window while the patient pushes into knee extension with the stimulator off, then the ramp, hold and ramp down at the current the assist rule set.
4. Model A labels each contraction. On fatigue the current is held; on guarding, stimulation stops and the clinician is notified. STOP stops stimulation at any time.
5. The brace sends the session summary; the patient rates comfort and pain in the app. A pain of 4/10 or more, or a pain-related STOP, blocks any increase until the clinician reviews.

Follow-up. The clinician reviews sessions remotely. After two weeks, Model C flags patients likely to stay below 110° of flexion at week 6. Standard rehabilitation, including walking, continues alongside.

VII.3 Usage limits and plan validity

A limit that exists only as an instruction is not a control. The firmware enforces these (D-06, D-12, D-11):

- **At most 2 sessions a day**, at least 3 h apart, using a stored counter and a sanity-checked clock. An invalid clock refuses the session and asks for a clinic reset.
- **Plan validity** of 36 h by default (clinician range 12–168 h). The second session of a day never exceeds the first session's ceiling.
- **Plan integrity.** Each plan carries a patient ID, an increasing version number, an expiry and a CRC-32. The brace rejects a corrupt, wrong-patient, expired or replayed plan and keeps the last valid one. An HMAC with a per-brace key follows after the MVP; CRC detects corruption but not tampering.

DECISION NEEDED · D-06 · Approval cadence

Recommended: one approval a day, 36 h validity, as above. Alternative: per-session approval as in v2, or auto-renewal after stable weeks (after the MVP).

VIII. AI personalisation layer

VIII.1 Models and results

The AI only suggests. Before each day's sessions it proposes three parameters, explains each, and the clinician accepts or overrides them one by one. Every result below is from a physiology-based simulator, not from patients **S Simulated**.

Model	When and where	Output	Simulated result
A. Muscle-state classifier (decision tree)	Every contraction, on the controller	Under-activated, on target, fatigued or guarding	93.5 % accuracy
B. Prescription recommender (3 random forests)	Before each day, on the backend	Ceiling, OFF time, contractions, each with two reasons	Ceiling within 5 mA 97 % (fixed protocol 56 %); OFF exact 86 %; contractions exact 93 %
C. Recovery prediction (random forest)	After 2 weeks, on the backend	Flexion at week 6; alert if below 110°	Error 2.2°; alert correct 90 %

Model A's labels are advisory: STOP and the current rules stay primary (SR-12). One caution on Model A: its inputs include EMG variability and median-frequency drop, so it must be given the same gated, valid-window features at run time that it saw in training; T-AI-06 checks that on real, unstimulated features.

VIII.2 Inputs, explanations, rules and learning

- **Inputs.** Model A uses five features per contraction (activation over target, median-frequency drop, EMG variability, angle reached, jerk). Model B uses 17 values from the last session: last plan, comfort, pain, flexion and its trend, activation and its trend, contraction level, fatigue, recovery after rest, Model A fractions, adherence and day.
- **Explanations.** Each suggestion is split into the contribution of every input along the trees' decision paths. The two strongest are shown as the reason, for example "comfort rating 3/3, day 29".

- **Rules before the clinician sees a value.** Ceiling rises by at most 10 mA per session and falls by 5 mA after pain of 4/10 or more or guarding above 30 %; then the cap of 50 mA applies. The threshold of 4/10 is a design value to confirm with the clinical adviser.
- **Clinician validation and learning.** Every decision is stored and used to retrain Model B. In simulation, a clinician with clinic-specific habits accepted all three values in 64 % of sessions before retraining and 92 % after **S Simulated**.

VIII.3 Data and deployment

No TKA dataset exists yet, so a physiology-based simulator generates virtual patients over 6 weeks, 2 sessions a day, with hidden traits (inhibition, recovery speed, fatigability, rest time, tolerance). Against a fixed protocol, the AI plus clinician cut sessions rated painful from 47 % to 1 %, with similar activation at week 6 (85 % versus 83 %). This checks the loop, not clinical efficacy: the simulator and the models were built by the same team, so agreement with itself is not evidence about patients.

The models are trained in Colab. Model A runs on the controller as generated C code. Model B is exported to JSON and runs in the Node.js backend; the export matches the Python model on 300 of 300 test cases, the one requirement verified by a recorded test **V Verified**.

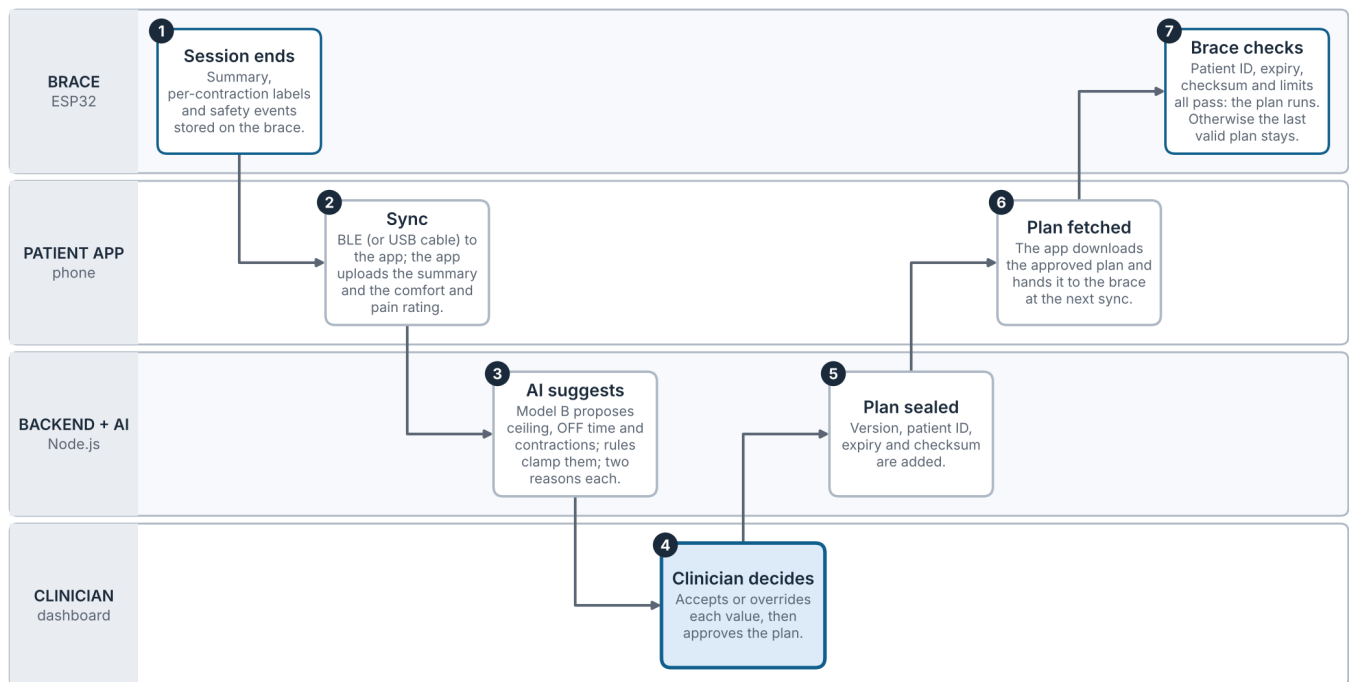
IX. Monitored parameters and interfaces

IX.1 Patient app and clinician dashboard

What drives the real-time loop stays on the brace. What people need to decide is shown as session-level values, without raw waveforms or pulse-level data.

- **Patient app (React Native).** The day's plan (contractions, ON and rest time, maximum mA) and whether it is approved, with its expiry; comfort (0–3) and pain (0–10) after each session; knee bend versus goal, a muscle engagement score, adherence over 2 sessions a day and weekly progress; STOP and emergency guidance. Patient-facing text sits in a string table (D-10): English and French for the demonstration, Arabic after the MVP.
- **Clinician dashboard (web and mobile).** A next-session panel with, for each parameter, the last value, the AI suggestion, its reasons and an accept or override choice, plus safety notes and an approve button. Fixed parameters are read-only. Charts of ROM, activation, fatigue, approved ceiling versus peak delivered; last sessions with comfort and pain; the decision log; the floor fraction and A_target as plan fields.

IX.2 Data flow and plan integrity



Safety net: the firmware refuses any value outside the locked limits, whatever the plan or the AI says.

No connection or no new approval: the brace keeps running the last approved plan until its expiry (default 36 h), then refuses to start.

Model C (recovery alert) runs on the same path after week 2 and only informs the clinician.

Figure 8. Data flow: brace → phone (BLE, or USB-serial fallback) → backend → AI suggestion → clinician approval → approved plan read by the brace at the next sync.

The brace never trusts a plan because of where it came from: the integrity checks of VII.3 run on every plan. Offline, the brace keeps working on the last valid plan until it expires; BLE loss mid-session changes nothing in the session, and the log is buffered. The BLE link is unfinished, so the end-to-end flow is not yet tested **G Gap**; the USB-serial fallback (D-08) is the schedule protection.

IX.3 Security and data handling

Role-based access, hashed passwords and expiring tokens are implemented in the local backend. Uploads with a plateau above 50 mA are rejected. Storage is a local database with anonymised patient IDs for the MVP (D-09); no personal data goes into logs. A hosted database, encrypted transport and a security review are not part of the MVP **G Gap**. The claim register (B.8) therefore does not allow “secure” or “GDPR-compliant”.

X. Safety architecture

X.1 Independent safety layers

Safety is enforced by independent layers, so no single algorithmic failure can produce unsafe stimulation, and every stimulation-path risk has a hardware control. Every layer is implemented except where marked; none has been verified on the assembled brace yet.

Layer	Threshold	Action	Status
1. Parameter limits	50 Hz; 150–400 μ s; ramps 3 s and 2 s; hold $\leq$ 15 s; OFF $\geq$ 30 s	Command refused, last valid plan kept	I Implemented
2. Approved ceiling	10–50 mA, approved per plan; floor $\leq$ ceiling	Command clamped	I Implemented
3. AI suggestion rules	+10 mA at most per session; –5 mA after pain or guarding above 30 %	Suggestion clamped, reason shown	I Implemented
4. Current tracking	Error above 20 % or 5 mA for 3 pulses	Burst stopped, clinician alerted	I Implemented

Layer	Threshold	Action	Status
5. Over-current	Above 60 mA in firmware; above 65 mA by the hardware comparator	SAFE_STOP; enable cut in under 1 ms	I Implemented
6. Lead-off and open circuit	Current below 10 % of command; noise floor outside window above 1 s	Stimulation stopped, clinician notified	I Implemented
7. Battery, USB, watchdog	Below 3.3 V; USB present; no heartbeat for 100 ms	SAFE_STOP, charge-only or reset	I Implemented
8. Stale data and sensor fault	Any sensor older than 500 ms	SAFE_STOP	D Planned
9. Usage and plan limits	2 sessions a day, 3 h gap, 36 h validity, plan integrity	Session refused	D Planned
10. Human oversight	Every plan approved by a clinician	Never an unapproved AI value	I Implemented
STOP button	Latching, within one-handed reach	Cut in under 10 ms, latched until restart	I Implemented
Skin-contact temperature	At or below 43 °C in the worst-case session	Design check	D Planned

X.2 Risk analysis (FMEA)

Ratings are 1–5 after mitigation: S severity, O occurrence, D detection difficulty; $RPN = S \times O \times D$. The first eight rows are the team's Phase 2 ratings, kept as they were. Rows F-09 to F-19 are proposed here and are the author's estimates for the team to review, not agreed ratings. Ratings are judgement, not measurement.

ID	Failure mode	Effect	Main mitigation	S	O	D	RPN	Origin
F-01	Over-current	Burn, pain	Approved ceiling, current tracking, 60 mA cut, 65 mA hardware trip	5	1	1	5	Team
F-02	Electrical fault or USB during stimulation	Shock	Battery-only stimulation, isolated control lines, USB lockout	5	1	2	10	Team
F-03	Firmware hang during a burst	Uncontrolled output	Separate real-time core, hardware watchdog	5	1	2	10	Team
F-04	AI ceiling above tolerance	Pain	+10 mA and pain rules, clinician approval	3	2	1	6	Team
F-05	Electrode detached	Current concentration	Open-circuit detection, stop and electrode check	4	2	1	8	Team
F-06	Wrong EMG reading or missed guarding	Dose too low or too high within the window	Lead-off check, artefact rejection, Model A, STOP button	3	3	2	18	Team
F-07	Potentiometer misalignment	Wrong angle	Close-fitting hinge, IMU cross-check, goniometer check	2	3	2	12	Team
F-08	Battery, skin or data issues	Thermal event, irritation, data loss	Protected cell, charge-balanced pulses, visible sites, hashed passwords	4	1	2	8	Team
F-09	EMG contaminated by stimulation or motion	Assist current wrong inside the floor–ceiling window	ATTEMPT state with stimulator off, posture and motion gate, floor and ceiling bound the result, Mode 0 fallback	3	3	2	18	New
F-10	Stale, tampered or wrong-patient plan applied	Wrong dose	Patient ID, version, expiry, checksum; limits in firmware; clinician approval; log	4	2	2	16	New
F-11	Contraction started in the wrong posture	Injury, fall, invalid measurement	Posture and motion gate before every contraction, seated-only instruction, STOP	4	2	1	8	New
F-12	More sessions than prescribed	Overdose, fatigue	Usage limiter in firmware	3	2	1	6	New
F-13	Sensor fault not detected	Control acts on frozen data	Stale-data timeout, Mode 0 fallback or stop	3	2	2	12	New
F-14	Module or battery overheats in the brace	Skin burn	Protected cell, thermal test, surface-temperature limit	4	1	2	8	New

ID	Failure mode	Effect	Main mitigation	S	O	D	RPN	Origin
F-15	Skin irritation or burn under the electrodes	Pain, skin damage	Electrode size chosen for current density, skin check before use, 50 mA cap, STOP	3	2	2	12	New
F-16	Clinician approves without review	AI value accepted unchecked	Two reasons shown, per-parameter accept, clamps, decision log audit	3	2	3	18	New
F-17	Use without clearance or with a contraindication	Serious harm	First session at the clinic, checklist before enrolment, surgeon clearance	5	1	3	15	New
F-18	Deep vein thrombosis symptoms missed	Delayed care	Instruction to stop and seek care for calf swelling or pain, reminder in the app	5	1	3	15	New
F-19	Connectivity loss blocks plan updates	Stale plan or interrupted therapy	Plan expiry, last valid plan, app alert	2	3	1	6	New

With the new rows, the highest RPN values are 18 for F-06 (wrong EMG reading), F-09 (EMG contaminated by stimulation or motion) and F-16 (clinician approves without review), then 16 for F-10 (stale or wrong-patient plan) and 15 for F-17 and F-18 (use with a contraindication; missed thrombosis symptoms). F-17, use without clearance or with a contraindication, has severity 5 and is controlled by procedure, not by the device, so it has no hardware layer: it relies on the clinic's checklist.

X.3 Eligibility, safe operation and emergency procedures

Contraindications (one list). Pacemaker or implanted stimulator; active deep vein thrombosis; damaged skin, wound or infection under the electrodes; impaired circulation; malignancy in the treated area; uncontrolled epilepsy; inability to use the STOP button; no surgeon clearance (use only from 48 h after surgery). **Precautions,** to be reviewed by the clinician before enrolment: sensory loss at the thigh, pregnancy, skin allergy to the electrodes, cognitive impairment that prevents safe use at home.

Topic	Rule
Potential risks	Burn or pain from excess current (ceiling, tracking, 60 mA cut, 65 mA trip). Electric shock (battery-only, isolated lines, blocked while charging). Software failure (hardware watchdog). Detached electrode (detected, stops). AI intensity too high (+10 mA per session, lowered after pain, clinician approval).
User controls	Red STOP button cuts stimulation in under 10 ms. Comfort (0–3) and pain (0–10) rating after each session. The clinician approves or overrides every parameter; nothing unapproved is applied.
Safe operation	First session at the clinic with the physiotherapist. Check contraindications first. Sit with the knee at 60°. Check the skin and electrodes before use. Unplug the charger and let the self-test finish. Maximum 2 sessions a day (the brace enforces it), approved plan only.
Emergency: pain or burning	Press STOP, remove the electrodes, report it in the app, contact the physiotherapist.
Emergency: calf swelling or pain	Stop and see a doctor urgently (possible thrombosis).
Emergency: chest pain or dizziness	Press STOP and call emergency services.
Emergency: smoke, heat or a device fault	Press STOP, switch off, do not reuse until checked.

X.4 Fault response

Every fault in B.6 has a defined response, a time limit and a test. In summary: faults that can cause harm go to SAFE_STOP in hardware or firmware; faults that only reduce control quality (invalid EMG, still flag false) degrade to Mode 0 at the floor; faults in plan, usage or clock refuse the session. A degraded session is logged as degraded and shown to the clinician.

X.5 Standards and regulatory position

The design is made with reference to IEC 60601-1 (basic safety), IEC 60601-1-11 (home use), IEC 60601-1-2 (EMC), IEC 60601-2-10 (nerve and muscle stimulators), IEC 62304 (software life cycle), IEC 62366-1 (usability), ISO 14971:2019 (risk management), ISO 10993-1 (biocompatibility of skin-contact parts), IEC 62133-2 (lithium cells) and ISO 13485 (quality management). **The device does not conform to any of them, and no conformity is claimed.** The editions and clauses were not checked for this document (Appendix G).

A device that delivers electrical stimulation for therapy would be regulated as a medical device. In the EU it would likely be class IIa or higher (MDR, rules 9 and 11); in the US a powered muscle stimulator is regulated under 21 CFR 890.5850. These classifications are the author's reading and must be confirmed with a regulatory adviser (SY-05). The stimulation path is verified on a resistive dummy load against the fault scenarios of B.6 before any use on a person.

X.6 Limitations

- The AI is trained and tested on simulated data only. It has not seen a real patient or a real clinician decision.
- No stimulation has been applied to a person with this device. The NMES module has been tested on a dummy load only.
- Stimulation on top of voluntary effort differs from the published protocols, which stimulate a resting muscle. In Mode 1 the patient pushes while the current is applied, which is not clinically validated.
- The controller does not measure force. It cannot show that a contraction reached a target torque.
- EMG is sensitive to motion artefact and electrode placement. The EMG interference test T-SEN-04a has not been run.
- The BLE link is unfinished; the end-to-end flow is untested. Data is local with basic security.
- Approval once a day is still clinician workload; whether it is acceptable is untested.
- The device is a prototype with no medical certification. Safety layers are implemented but fault injection on the assembled device has not been run.

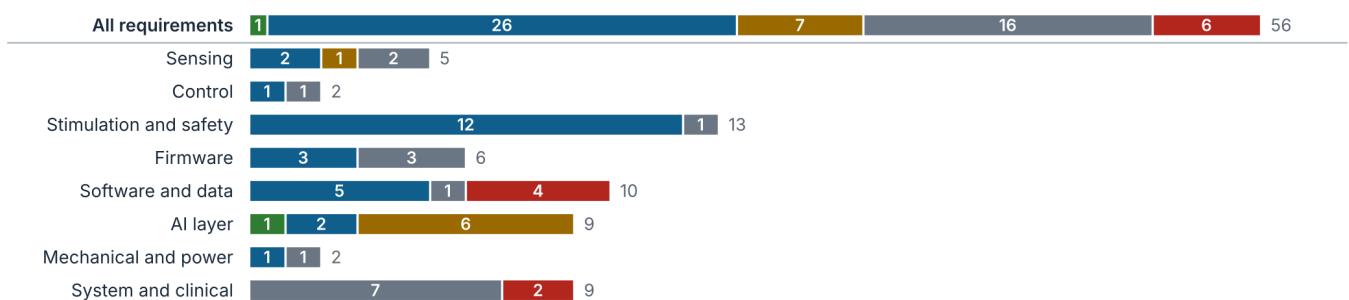
XI. Verification and roadmap

XI.1 Where the evidence stands

Of 56 requirements, 1 is verified, 26 are implemented and unverified, 7 are simulated only, 16 are planned and 6 are gaps. The shape of Figure 9 is the honest summary: much has been built, little has been proven. "Implemented" includes tests the team reports (EMG on a muscle, NMES on a dummy load, angle filter tuning) until their records are checked against the pass criteria in B.5.

■ V Verified ■ I Implemented ■ S Simulated ■ D Planned ■ G Gap

Requirements by subsystem and evidence status (counts)



Test catalogue by priority (counts)



Figure 9. Requirements by subsystem and evidence status, and the test catalogue by priority. Both are generated from the same master list (Appendix A).

XI.2 Verification approach

Verification is organised by risk, in this order: safety chain on the bench (T-HW), firmware logic (T-FW), sensing on volunteers without stimulation (T-SEN), software flow (T-SW), AI checks (T-AI), then integrated system tests on a dummy load and the demonstration rehearsals (T-SYS). Part B defines each of the 51 tests with a method, a pass criterion and the evidence to keep, and a fault-injection matrix for the assembled device.

No test in the MVP applies stimulation to a person. The evoked-response test T-SEN-04b, which decides the guard time, needs that and waits for ethics approval.

XI.3 Roadmap after the MVP

Step	What	Needs
1	Finish BLE; end-to-end test; hosted database and security review	Software time; hosting; security review
2	Record sessions on healthy volunteers (no stimulation); retrain Models A and B on real features	Institutional approval, consent form
3	Load cell and HX711; EMG-versus-force study; Mode 2 and extremum-seeking tuning as a supervised commissioning aid	Load cell (D-04); bench study; ethics approval
4	Evoked-response test and guard-time measurement on a person (T-SEN-04b)	Ethics approval, clinical supervision, passed T-HW-01 to 11
5	Plan HMAC, auto-renewal rules, Arabic interface	Firmware and app work
6	Ethics-approved pilot with TKA patients; regulatory pathway and technical file	Clinical partners; regulatory adviser

XII. Conclusion

The smart knee wearable aims to close the gap between devices that sense and devices that treat. In v3.0 the controller is honest about what its sensors can support: voluntary EMG is read with the stimulator off and sets a bounded dose once per contraction; a hardware-backed safety chain, a fixed-dose fallback and clinician approval surround it; and an AI layer prepares each day's plan from the patient's own data. Compared with Phase 1, the prototype has a printed orthosis, complete firmware, tested sensing and stimulation modules, an explainable AI validated in simulation, and working patient and clinician interfaces. The next phase is verification: the safety chain first, then the integrated system on a dummy load.

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- Standards listed in X.5 (editions to be confirmed). Further sources and verification notes: Appendix G.
- Team. Technical Architecture Report (Phase 1), SymbioMed Challenge, IEEE TRSYP 3.0, 2026.

PART B · FINAL MVP GUIDELINES

B.1 MVP definition and success criteria

Part B turns the specification into a plan the team can finish, defend and demonstrate. The MVP is defined by what a jury can see working, what the team can show evidence for, and what the team refuses to claim.

RULE · The MVP in one sentence

A brace that accepts an approved, integrity-checked plan, runs a session on a dummy load with the hardware safety chain proven, uses voluntary EMG from a volunteer (no stimulation on anyone) to set the plateau current within the plan's floor and ceiling, logs the session, and shows it in the app and dashboard.

Success criteria

#	Criterion	Evidence	Tests
1	Safety chain proven on the assembled device: STOP, USB lockout, both over-current cuts, watchdog, open circuit, battery, boot state	Scope captures and logs for T-HW-03 to 10	T-HW-03 ... 10
2	Stimulation waveform and current within tolerance into a dummy load over 10–50 mA	Scope captures, CSV	T-HW-01, 02
3	A complete Mode 0 session on a dummy load with a clean log	Log, video	T-SYS-01
4	EMG from a volunteer is clean, repeatable and not corrupted by the NMES module running into the dummy load	Recordings, SNR table, interference table	T-SEN-01, 03, 04a
5	Mode 1 changes the plateau current in the expected direction within the floor and ceiling, with fallback to Mode 0 on every invalid input	Log, video, unit-test report	T-FW-07, T-SYS-02
6	Plan travels from dashboard to brace (BLE or USB-serial) and the session log returns	Video, backend log	T-SW-01, 02, 04
7	Fault-injection campaign on the assembled device passes	Filled matrix B.6	T-SYS-03
8	Demonstration runs three times in a row without intervention	Rehearsal records	T-SYS-06
9	Every claim made in the demonstration appears in the claims register with its evidence	B.8	—
10	Cost is at or below 575 TND with the optional items shown outside the cap	Appendix B tracker	—

What the MVP is not

- Not a clinical study and not a test on any person with stimulation.
- Not a force-controlled or self-tuning device. The MVP controller is a bounded assist rule; extremum seeking is a roadmap study.
- Not secure data hosting, and not a certified or certifiable design.

B.2 Scope: must, should, could, won't

Everything in the specification falls into one of four groups. The cut-line in B.9 says what to drop first if time runs short; the won't-have list is how scope creep (P-08) is refused.

Group	Items
Must (the MVP fails without it)	Safety chain and its tests (T-HW-01 to 10); parameter limits, state machine and timing (T-FW-01 to 05, 07); Mode 0 session on a dummy load; EMG baseline, reference MVC and interference test (T-SEN-01, 03, 04a); gate thresholds (T-SEN-05); plan approval and integrity, dashboard to brace in either transport (T-SW-01 to 04); Model A on the device and Model B export parity (T-AI-01 to 03); fault-injection campaign; demonstration rehearsals; claims register; cost tracker.
Should (strengthens the case)	Mode 1 closed-loop demonstration; isolation, thermal and battery checks; usage limiter; logging integrity; offline behaviour; basic security checks; dashboard correctness; fatigue index and repetition count on volunteers; usability session; soak test; runtime test; fit and hinge cycling.
Could (if time remains)	Real-feature dry run of Model A (T-AI-06); Arabic strings (D-10); second IMU on the thigh (D-05); a load cell mounting point on the shell (D-04, mount only).

Group	Items
Won't (this phase)	Stimulation on any person; Mode 2, extremum-seeking tuning and force sensing; evoked-response listening gaps (T-SEN-04b); hosted database and encrypted transport; plan HMAC; auto-renewal of plans; regulatory submission; clinical claims.

B.3 Configuration baseline, change control and regression

A prototype that changes in the last week cannot be trusted in the last day. These rules keep the evidence valid (P-13).

1. **Baseline tag.** Tag firmware, app, backend, models, plan format and the parameter register together at design freeze (week -4). Every log record carries the firmware version.
2. **Freeze dates.** Parameters and hardware at week -4; firmware features at week -3; app and backend at week -2; only fixes after that, each with a recorded reason.
3. **Change record.** Every change after the baseline needs: what, why, who, the tests it forces to be re-run (table below), and the new tag.
4. **Evidence belongs to a tag.** A test record names the tag it ran on. A record from an older tag is kept but does not count for a changed area.

If this changes	Re-run at least
Hardware safety chain, shunt, comparator, enable gate, watchdog	T-HW-02 to 10, T-SYS-03
Current limits, ceiling, ramps, pulse width, frequency	T-HW-01, 02, T-FW-01, 04, 07
State machine or timing	T-FW-02, 03, T-SYS-01, 03
Plan format, integrity or usage limits	T-FW-05, 06, T-SW-02, 03, 04
Assist rule or its gains	T-FW-07, T-SYS-02
Gate thresholds, EMG filters or normalisation	T-SEN-01, 03, 04a, 05, T-FW-07
Model A or the feature code	T-AI-01, 06
Model B, export or suggestion rules	T-AI-02, 03, T-SW-04
App or backend	T-SW-01 to 04, 07
Battery, charger or enclosure	T-HW-06, 09, 12, 13, T-PWR-01, T-MECH-01

B.4 Build and integration plan

Weeks are counted back from the demonstration (week 0). The date of the demonstration was not given; the team should write it against week 0. The order puts safety evidence first, because everything else is built on it, and puts the schedule-risk items (BLE, EMG interference) early. Owners are to be named by the team.

Week	Work	Gate (all must pass to move on)
-6	Resolve D-01, D-02, D-07. Order spares and the ADS1115. Freeze the parameter register. Start the institutional approval and consent form for volunteer recordings. Build the dummy-load fixtures. Start BLE or USB-serial.	D-01, D-02, D-07 recorded in B.10. Orders placed.
-5	T-HW-01, 02, 03, 06, 07 on the bench. T-FW-01 and 02. First volunteer EMG baseline (T-SEN-01). Angle calibration (T-SEN-02).	STOP, USB lockout and watchdog pass. If any fails, stop other work on it (P-03).
-4	T-HW-04, 05, 08, 09, 10. T-SEN-04a EMG interference. T-SEN-03 reference MVC. Design freeze for hardware and parameters.	All of T-HW-01 to 10 pass. T-SEN-04a result decides Mode 1 (pass) or Mode 0 only (fail).
-3	T-FW-03, 04, 05, 07. Gate thresholds (T-SEN-05). Plan integrity end to end. BLE or USB-serial working. Firmware feature freeze.	T-SW-01 passing, or USB-serial chosen (P-01 trigger).
-2	T-SYS-01 Mode 0 session. T-SYS-02 Mode 1 demonstration on a dummy load. T-AI-01 to 03. T-SW-02 to 04. App and backend freeze.	Mode 0 session clean; Mode 1 either passes or is dropped from the demo.
-1	T-SYS-03 fault-injection campaign on the assembled device. Soak. Evidence pack assembled. Demonstration rehearsal 1 and 2.	Campaign passes. Claims register matches the evidence.

Week	Work	Gate (all must pass to move on)
0	Rehearsal 3 on the morning. Demonstration. Backups ready.	Three clean rehearsals in a row (T-SYS-06).

Two things are not scheduled because they depend on others: the institutional approval for volunteer recordings (P-06) and delivery of parts (P-04). If either is late, the plan keeps the safety and bench work and drops the volunteer-dependent items in the order of B.9.

B.5 Test catalogue

Each test has a method, a pass criterion and the evidence to keep. Priority is Must, Should, Could or After MVP. "Volunteer (no stimulation)" means a healthy adult records EMG or angle with no current applied to them. The pass criteria are proposed values for the team to confirm; where a number is an engineering choice and not a datasheet value, it is a design value. All stimulation tests use a resistive dummy load (500 Ω , 1 k Ω , 2 k Ω , power rating and voltage rating suited to the current) and never a person.

CAUTION · Dummy-load safety

Before any stimulation test: battery-only, brace off any person, load connected, a second person present for the STOP tests, and a written record of the setup. Check the load's power rating: 50 mA into 2 k Ω is 5 W.

B.5.1 Hardware and safety chain (T-HW)

Test	Method	Pass criterion	Evidence
T-HW-01 Must	NMES waveform fidelity Set 50 Hz, 250 μ s, 30 mA into 1 k Ω . Capture at least 10 pulses on a scope across the load. Repeat at 150 and 400 μ s.	Frequency 50 Hz $\pm$ 2 %; pulse width $\pm$ 10 %; positive and negative phase charge within $\pm$ 5 %; no DC offset above 2 % of peak.	Scope captures, CSV, setup photo Covers: CF1, CF8
T-HW-02 Must	Current accuracy and linearity Command 10, 20, 30, 40 and 50 mA into 500 Ω , 1 k Ω and 2 k Ω . Measure across a sense resistor or with a calibrated meter.	Within $\pm$ 10 % or $\pm$ 2 mA, whichever is larger, at 500 Ω and 1 k Ω . At 2 k Ω , record the compliance limit.	Table and plot, instrument IDs Covers: MF4, SR-02, SR-04
T-HW-03 Must	STOP latency and latch Press STOP 20 times during ramp-up, hold and ramp-down at 30 and 50 mA into 1 k Ω . Trigger the scope on the button line.	Current below 5 % of plateau within 10 ms in 20 of 20 presses; no restart on release; restart only by explicit reset.	Scope captures, min, mean and max table Covers: CF3
T-HW-04 Must	Hardware over-current trip, independent of firmware Hold the ESP32 in reset. Raise the sensed current with a bench source through the sense path until the comparator trips.	Enable falls within 1 ms of the current crossing 65 mA $\pm$ 5 %; the trip is latched until restart.	Scope captures at three sweep rates Covers: SR-05
T-HW-05 Must	Firmware over-current cut Inject a sense voltage equal to 59 mA and to 61 mA in a test build.	61 mA gives SAFE_STOP within one supervision interval (20 ms or less); 59 mA does not; 10 of 10 each.	Serial log and scope capture Covers: SR-05
T-HW-06 Must	USB lockout Plug the USB-C charger before a session and during a session.	Before: session refused with a message. During: output off within 10 ms and stays off until the cable is removed and the brace restarted. Charging works. 10 of 10.	Scope captures, log Covers: SR-07
T-HW-07 Must	Watchdog Insert an infinite loop in the real-time task of a test build during stimulation. Repeat with a brown-out.	Output current zero within 120 ms of the last heartbeat; 10 of 10.	Scope captures Covers: SR-08
T-HW-08 Must	Open circuit and lead-off Disconnect the load during the hold, then reconnect. Repeat with a 10 k Ω load.	SAFE_STOP within 3 pulses (60 ms); no restart on reconnect; the log shows the cause; the 10 k Ω case is flagged.	Scope captures, log Covers: SR-06, CF2
T-HW-09 Must	Battery undervoltage Sweep a bench supply from 4.2 V to 3.0 V at 10 mV/s during a session.	SAFE_STOP at 3.3 V $\pm$ 0.05 V before the ESP32 brown-out threshold; no spurious stimulation while the supply falls.	Supply log, scope capture Covers: SR-09

Test	Method	Pass criterion	Evidence
T-HW-10 Must	Boot and reset state Run 50 power cycles, 50 resets, 10 flash cycles and 10 brown-outs at different supply levels with the scope on the load.	Output current is zero in every case (enable pulled down).	Scope captures, run sheet Covers: SR-09
T-HW-11 Should	Isolation check With the battery removed, measure insulation between the control side and the stimulation side of each optocoupler at 100 V DC.	At least 10 MΩ and no breakdown (design target).	Meter readings, photo Covers: SR-10
T-HW-12 Should	Thermal check Run the worst-case plan (50 mA, 400 μs, 20 contractions, OFF 30 s) on a dummy load inside the closed brace. Log temperatures with thermocouples.	Skin-contact surfaces at or below 43 °C; no component above its rating.	Temperature log, photo Covers: SR-13, CF8
T-HW-13 Should	Battery and charging check Charge and discharge the protected cell with a bench supply and electronic load. Check termination and protection cut-offs.	Charge ends at 4.20 V ± 0.05 V; protection cut-offs verified; case temperature below 45 °C.	Logs, photo Covers: CF8, CF2

B.5.2 Firmware (T-FW)

Test	Method	Pass criterion	Evidence
T-FW-01 Must	Parameter validation A script sends 1000 plans with random and boundary values for every parameter, including expiry and floor.	All out-of-range plans refused, all in-range plans accepted, last valid plan unchanged, no crash.	Script, log Covers: CF1, SR-01
T-FW-02 Must	State machine coverage Drive every transition and inject every fault listed in B.6.	All transitions covered; every fault ends in SAFE_STOP with a cause code; SAFE_STOP exits only by a power cycle.	Coverage table, log Covers: FW-01
T-FW-03 Must	Real-time timing Log the control-loop period for 60 min and 10 sessions while BLE and logging are active.	Period 100 ms ± 5 ms at the 99.9th percentile; no overrun above 20 ms; no watchdog trip; core-0 load below 50 %.	Timing histogram, CPU log Covers: FW-02
T-FW-04 Must	Current supervision logic Inject sense deviations just below and above 20 % and 5 mA for 1, 2, 3 and 4 pulses.	Burst stops on the third consecutive violating pulse and not on the second.	Log Covers: SR-04
T-FW-05 Must	Plan store and integrity Cut power during plan writes (100 trials). Send a corrupt checksum, a wrong patient ID, an expired plan and a replayed older version.	Only a complete, valid, newer plan is accepted; otherwise the last valid plan is kept; an expired plan refuses to start with a message.	Log, test script Covers: CF4, MF10, FW-03
T-FW-06 Should	Usage limiter Try a third session in a day, a second session before the minimum gap, and a manipulated clock.	All refused; an invalid clock refuses the session and asks for a clinic reset.	Log Covers: MF10
T-FW-07 Must	Assist rule unit tests Run 200 scripted A_v sequences (zero effort, above target, invalid, saturated) on a PC build and on the brace.	Command always between floor and ceiling; change per contraction at most 3 mA; integral bounded; invalid input holds the last value; matches the Python reference within 0.1 mA.	Test report, reference script Covers: MF4, FW-05, SR-02
T-FW-08 Should	Logging integrity Log 100 sessions with power cycles in between.	No lost or corrupt record; time stamps increase; firmware version recorded in every record.	Log dump Covers: FW-04

B.5.3 Sensing (T-SEN)

Test	Method	Pass criterion	Evidence
T-SEN-01 Must	EMG baseline and signal-to-noise On 5 volunteers, record 30 s rest, 3 maximal contractions of 5 s and 3 holds at 30 % of maximum.	Rest RMS at most 10 % of maximal RMS; no saturation; median frequency computed in every valid window.	Recordings, plots, consent record Covers: MF1, SY-03

Test	Method	Pass criterion	Evidence
T-SEN-02 Must	Knee angle accuracy Use a jig at 0°, 30°, 60°, 90° and 120°, three repeats, three brace positions. Repeat after 1000 hinge cycles.	Error at most 3° everywhere and at most 2° at 60° ± 10°; drift at most 1° over 30 min and after 1000 cycles.	Table, jig photo Covers: MF2
T-SEN-03 Must	Reference MVC repeatability One volunteer, brace fitted and removed on three occasions, three maximal efforts each, then a 30 % target hold with visual feedback.	Coefficient of variation of the reference at most 15 %; the 30 % hold reproduced within ± 5 % of the reference.	Recordings, table Covers: MF1
T-SEN-04a Must	EMG interference from the NMES module EMG module on a real muscle while the NMES module is idle, enabled at 0 mA, and stimulating into the dummy load at 20, 40 and 50 mA.	Idle and 0 mA: noise within 10 % of baseline and A_v within ± 5 % of the no-module condition. Stimulating: added noise and any saturation are reported.	Recordings, table Covers: MF1
T-SEN-04b After MVP	Evoked response and recovery time Needs stimulation on a person with ethics approval. Measure how long the evoked response and artefact last after the current ends.	Gives the guard time and decides whether listening gaps can ever be used.	Study report Covers: MF1, MF8
T-SEN-05 Must	Gate thresholds 5 volunteers sit still at 60°, make small movements, swing the leg, stand and sit, tap the strap, and hold 40° and 80°.	Gate rejects moving or off-posture windows in at least 98 % and accepts still 60° windows in at least 90 %; thresholds recorded.	Recordings, confusion table Covers: MF8, FW-06
T-SEN-06 Should	Fatigue index trend 5 volunteers hold 30–40 % of maximum until task failure or 3 min. Compute the median-frequency slope.	Slope negative in at least 80 % of runs; otherwise the fatigue claim is removed.	Recordings, plots Covers: MF3
T-SEN-07 Should	Repetition count and still flag Five runs of 20 repetitions.	Count error at most ± 1 in at least 95 % of runs; still flag within 1 s.	Run sheet Covers: MF2, MF8

B.5.4 Software and data (T-SW)

Test	Method	Pass criterion	Evidence
T-SW-01 Must	BLE link 100 connect, sync and disconnect cycles at 1 m and 5 m. Test the USB-serial fallback.	At least 99 of 100 syncs complete; an interrupted transfer fails safely; no stimulation change during sync.	Log Covers: MF7, IR-04
T-SW-02 Must	Approval workflow Scripted cases: accept all, override each parameter, reject.	The brace only ever receives approved values; the dashboard shows exactly the stored values; 10 of 10 cases.	Script, screenshots Covers: MF6, CF4, IR-02
T-SW-03 Must	Plan integrity (backend side) Upload malformed plans, values above the cap and plans for the wrong patient.	All rejected; uploads above 50 mA refused.	Log Covers: MF6, IR-05
T-SW-04 Must	End-to-end cycle 20 consecutive cycles: brace, app, backend, approval, brace.	20 of 20 complete; technical latency at most 10 s excluding the clinician; logs reconcile.	Log, video Covers: SY-01, MF7, IR-01, IR-03
T-SW-05 Should	Offline behaviour Switch off the phone, the backend and the network at each step.	The brace runs the last valid plan within its expiry; the app shows the state; no crash; data uploads when the link returns.	Log Covers: IR-05
T-SW-06 Should	Security basics Check token expiry, role separation, password hashing, input validation and that no secrets are in the repository.	Checklist 100 % complete; tokens expire within 24 h; a patient cannot call clinician endpoints.	Checklist Covers: CF9, IR-06
T-SW-07 Should	Dashboard correctness Sample 100 displayed values against the stored records.	100 of 100 match; charts match the raw logs.	Sample sheet Covers: IR-02

B.5.6 System (T-SYS)

Test	Method	Pass criterion	Evidence
T-SYS-01 Must	Integrated Mode 0 session on a dummy load 10 contractions at 30 mA, OFF 45 s, full brace, app and backend running.	All contractions delivered with timing 3 s, 15 s, 2 s and the approved OFF; current within $\pm 10\%$; log complete.	Log, scope captures Covers: MF4, CF1
T-SYS-02 Must	Mode 1 demonstration on a dummy load Volunteer pushes at about 0, 15, 30 and 50 % of maximum during the ATTEMPT state. Stimulation goes to the dummy load only.	Plateau current follows the rule within ± 1 mA; never below floor or above ceiling; change per contraction at most 3 mA; an invalid attempt holds or falls back to Mode 0.	Log, video Covers: MF4, MF9, FW-05, FW-06
T-SYS-03 Must	Fault-injection campaign Run every row of B.6, three repetitions each.	Expected response in 100 % of runs.	One record per row Covers: SY-02, SR-11, CF2
T-SYS-04 Should	Usability Three people who are not on the team fit the brace, start a session and press STOP on request.	3 of 3 finish without help in 5 min or less; STOP found within 3 s; errors logged.	Observation sheets, video Covers: CF10
T-SYS-05 Should	Soak 14 sessions over 7 days on the dummy load with app and backend.	No unexpected stop; battery log complete.	Logs Covers: FW-02
T-SYS-06 Must	Demonstration rehearsals Three consecutive full runs of the script in B.7.	3 of 3 without intervention, each within 7 min.	Videos, timing sheet Covers: MF9
T-PWR-01 Should	Battery runtime Two 40-min sessions at worst-case settings on a dummy load.	At least 20 % of the usable capacity remains.	Discharge log Covers: CF6
T-MECH-01 Should	Fit and access Fit the brace on at least 3 volunteers with different thigh and shank sizes, no stimulation, 40 min wear.	Fitted in 5 min or less; patella and electrode sites visible; STOP reachable one-handed when seated; no pinch points.	Photos, sheet Covers: CF5
T-MECH-02 Should	Hinge cycling 1000 flexion cycles from 0° to 120°.	Potentiometer drift at most 1°; no loose parts.	Log, photo Covers: CF5, MF2

B.6 Fault-injection matrix

Run on the assembled device on a dummy load (T-SYS-03). Each row is injected at least three times; record the measured response time against the limit. A response slower than its limit, or any unexpected current, is a failure of the safety chain and stops other work (P-03). Fault cases that need the brace on a person are excluded.

ID	Fault	How to inject	Expected response	Limit	Test	Result
FI-01	STOP pressed in any state	Press the button during ATTEMPT, ramp-up, hold, ramp-down and rest	Output off, SAFE_STOP latched, cause logged	10 ms	T-HW-03	[] pass [] fail
FI-02	USB plugged in during stimulation	Plug the charger into the running brace	Output off, charge-only, no restart until unplugged and restarted	10 ms	T-HW-06	[] pass [] fail
FI-03	Firmware hang on core 0	Test build with an infinite loop in the real-time task	Watchdog cuts the enable line, output zero	120 ms	T-HW-07	[] pass [] fail
FI-04	Core 1 stall, sensor data stops	Test build stops publishing EMG and angle	Stale-data detection, SAFE_STOP	500 ms	T-SYS-03	[] pass [] fail
FI-05	Over-current above 60 mA (simulated)	Inject a sense voltage equal to 61 mA	SAFE_STOP, cause logged	20 ms	T-HW-05	[] pass [] fail
FI-06	Over-current above 65 mA, MCU halted	Bench source through the sense path with the ESP32 in reset	Comparator trips, enable low, latched	1 ms	T-HW-04	[] pass [] fail
FI-07	Open circuit, electrode lifted	Disconnect the load during the hold	SAFE_STOP, no restart on reconnect	60 ms	T-HW-08	[] pass [] fail
FI-08	Output short	Short the dummy load	Module limits current, tracking error or over-current gives SAFE_STOP	60 ms	T-HW-08	[] pass [] fail

ID	Fault	How to inject	Expected response	Limit	Test	Result
FI-09	Current-sense line open or shorted	Disconnect or short the sense signal	Implausible reading gives SAFE_STOP (fail-safe)	60 ms	T-FW-04	[] pass [] fail
FI-10	Battery undervoltage	Bench supply sweeps down to 3.0 V	SAFE_STOP at 3.3 V, no spurious pulses	at 3.3 V	T-HW-09	[] pass [] fail
FI-11	Power cut and restore mid-contraction	Switch the brace off and on	Output zero, restart in BOOT then SELF_TEST, no automatic resume	immediate	T-HW-10	[] pass [] fail
FI-12	Corrupt plan	Send a plan with a bad checksum	Rejected, last valid plan kept	n/a	T-FW-05	[] pass [] fail
FI-13	Out-of-range plan	Send 80 mA, 500 μ s, OFF 10 s	Refused, reason logged	n/a	T-FW-01	[] pass [] fail
FI-14	Expired or replayed plan	Send an old version or an expired plan	Rejected, session refuses to start with a message	n/a	T-FW-05	[] pass [] fail
FI-15	Wrong patient ID	Send a plan for another patient	Rejected	n/a	T-FW-05	[] pass [] fail
FI-16	BLE disconnect mid-session	Switch the phone off during a session	Session continues locally, log buffered, no parameter change	n/a	T-SW-05	[] pass [] fail
FI-17	Potentiometer disconnected or out of range	Unplug or short the potentiometer	Posture gate fails, no start; during a session, SAFE_STOP	500 ms	T-SYS-03	[] pass [] fail
FI-18	IMU lost (I2C error)	Unplug the IMU	Still flag invalid, gate fails, current contraction ends and the session stops	500 ms	T-SYS-03	[] pass [] fail
FI-19	EMG lead-off, saturated or flat	Detach an EMG lead	EMG invalid, last A_v held, then Mode 0 at the floor; clinician notified after 3 invalid attempts	1 s	T-SYS-03	[] pass [] fail
FI-20	Shank moves during ATTEMPT	Move the leg	Attempt rejected and repeated up to 3 times, then Mode 0 for that contraction	window	T-SEN-05	[] pass [] fail
FI-21	Third session or too early a restart	Start sessions outside the limits	Refused with a message	n/a	T-FW-06	[] pass [] fail
FI-22	Invalid clock	Set the phone or brace clock wrongly	Session refused, clinic reset requested	n/a	T-FW-06	[] pass [] fail
FI-23	Backend unavailable	Stop the backend	App shows no new plan; brace uses the last valid plan until it expires	n/a	T-SW-05	[] pass [] fail

B.7 Safe demonstration plan

The demonstration shows the system working, shows how it stays safe, and states what has not been proven. It never stimulates a person. The brace sits on a stand or a model leg, with stimulation into a dummy load; a volunteer's EMG drives the command through the assist rule, which is why the demo can show Mode 1 without applying any current to them.

Set-up

- Brace on a stand or mannequin leg; dummy load (1 k Ω) connected to the stimulation leads, with a scope or a current display visible.
- Volunteer wears only the EMG electrodes on the thigh, on a second cable; no stimulation lead is connected to them.
- Laptop with dashboard and backend running locally; phone with the app; USB-serial cable as the plan transport fallback; spare laptop; pre-recorded video of a full run; printed evidence pack (B.8).
- Everything works without the venue network. Battery fully charged; a second charged brace battery.

Seven-minute script

Time	Show	Say (wording from the claims register)
0:00–0:45	The problem slide and the brace on the stand	"Quadriceps weakness after knee replacement is the barrier; today's units are open loop. This is a research prototype and no stimulation has been applied to a person."
0:45–1:45	Dashboard: the clinician sees the AI suggestion with two reasons, accepts one value, overrides another, approves	"The AI only suggests. Nothing reaches the brace without the clinician. Results shown for the AI are from simulated patients."
1:45–2:30	Plan reaches the brace; the brace rejects a corrupted plan; shows version and expiry	"The brace checks every plan and keeps the last valid one."
2:30–4:00	Session in Mode 1 into the dummy load: the volunteer pushes with the stimulator off; the plateau current changes; the scope shows the pulses	"The current is set once per contraction from voluntary EMG with the stimulator off, between a clinician floor and an approved ceiling. It does not measure force."
4:00–5:15	Safety: press STOP mid-hold and show the cut on the scope; plug the USB in; lift a load lead	"These protections work in hardware and are latched. Measured response times are in the evidence pack."
5:15–6:15	App: comfort and pain, session summary; dashboard chart update	"The loop is: suggest, approve, deliver, log, retrain."
6:15–7:00	Limitations and roadmap slide	"Not yet tested with stimulation on people; fault injection results are in the pack; next: volunteers, a force sensor, ethics-approved pilot."

Fallbacks

If this fails	Do this
BLE does not connect	Switch to the USB-serial plan transfer; say so
Backend or dashboard fails	Use the spare laptop; else the pre-recorded video, announced as a recording
EMG noisy or lead-off	The brace goes to Mode 0 at the floor by design; show that as the fallback working
Brace does not start	Show SAFE_STOP and the log, which is the correct response to a fault; use the recorded video for the session
Any unexpected output or heat	Press STOP, disconnect the battery, stop the demonstration, show the log

Questions to prepare for

Likely question	Honest answer
Has it been used on a patient or on you?	No. Stimulation has only been applied to a dummy load.
Is the 97 % real?	It is from simulated patients built by the team; it checks the loop, not clinical accuracy. It needs retraining on real clinician decisions.
Is it a closed-loop force controller?	No. It sets a bounded plateau once per contraction from voluntary EMG. Force is not measured.
Why not adjust the current while it flows, from EMG?	While current flows, EMG is mostly the evoked response, not the patient's effort.
What if the software crashes?	The watchdog, over-current trip, STOP and USB lockout work without the processor; here are the measured times.
Is it safe? Is it medical grade?	It is a research prototype. We say what we measured, not that it is safe for patients.
Which standards does it meet?	None are claimed. It is designed with reference to them; conformity needs a regulatory process.
Where does the data go?	Locally in the prototype, with anonymised IDs; a hosted, secured database is planned.
Why is the ceiling 50 mA when the module goes to 57.6?	To keep the margins between the approved ceiling, the module maximum, the firmware cut and the hardware trip.
How much did it cost?	The tracker total in Appendix B against 575 TND; optional items are shown outside the cap.

B.8 Evidence pack and claims register

The evidence pack is a folder, printed in summary and kept in two copies (P-14). Name every record 'T-ID_tag_date_initials' and keep it with its raw data.

Folder	Contents
01_baseline	Tags, parameter register, schematics, firmware version list, photos of the build
02_bench	T-HW and T-FW records: scope captures, CSV, logs, setup photos
03_sensing	T-SEN recordings, SNR and interference tables, calibration table, consent form
04_system	T-SW, T-AI, T-SYS records, videos, backend logs
05_faults	Filled B.6 matrix with measured response times
06_claims	Claims register, slides, script, the never-say list
07_cost	BOM tracker and receipts

Claims register. Say the “allowed now” wording. Upgrade it only when the named test has passed and its record is in the pack. Never say the words in the last column.

Topic	Allowed now	Upgrade after	Never say
Safety hardware	STOP, USB lockout, over-current trip, watchdog and enable gate are implemented.	“...and verified on a dummy load” after T-HW-03 to 10 pass.	Safe for patients; clinically safe; medical-grade
NMES output	A commercial constant-current module was tested on a dummy load.	“...output within $\pm 10\%$ of the set current” after T-HW-02 passes.	Certified output; compliant with IEC 60601-2-10
EMG sensing	A surface-EMG module was tested on a real muscle and gives RMS and median frequency.	“...reference repeatable within $\pm 5\%$ MVC” after T-SEN-03 passes.	Measures muscle force
Knee angle	Angle comes from a potentiometer and an IMU, with the filter tuned against a goniometer.	“...within 3° over $0-120^\circ$ ” after T-SEN-02 passes.	Clinical-grade goniometry
Control	The assist current is set once per contraction from EMG measured with the stimulator off, between a clinician floor and an approved ceiling. Force is not measured.	After T-SYS-02 passes on a dummy load, add “demonstrated on a dummy load with volunteer EMG”.	Closed-loop force control; adaptive PID that tunes itself
Self-tuning	Extremum-seeking gain tuning is a roadmap item studied in simulation.	Only after a force sensor, a bench study and ethics approval.	Self-tuning (as a feature)
AI performance	In simulation, Model B placed the ceiling within 5 mA in 97 % of sessions (fixed protocol 56 %). Simulated patients only.	Only after retraining on real clinician decisions.	The AI is 97 % accurate; the AI reduces pain
Human oversight	A clinician approves every plan; the firmware enforces the limits whatever the plan says.	—	The AI controls stimulation; fully autonomous
Clinical benefit	NMES after TKA follows a published protocol (Stevens-Lapsley et al., 2012). This device has not been tested clinically.	Only after an approved clinical study.	Improves recovery; clinically validated; reduces pain
Testing on people	No stimulation has been applied to a person with this device.	Update after an approved study.	Tested on patients
Regulatory status	Research prototype, not certified.	Update after a regulatory review.	CE marked; FDA cleared; compliant with IEC 60601
Data	Data is stored locally in the prototype; a hosted database is planned.	After T-SW-06 and a security review.	Secure; encrypted; GDPR-compliant
Cost	Prototype bill of materials of X TND against the 575 TND cap (fill in from Appendix B).	—	A figure that is not in the tracker

B.9 Project risks and cut-line

ID	Risk	L	I	Mitigation	Trigger
P-01	BLE link not finished in time	M	H	USB-serial fallback (D-08); freeze BLE scope; test every week	T-SW-01 not passing at week -3: demonstrate over the cable path
P-02	EMG contaminated by the NMES module or by motion	M	H	ATTEMPT design; run T-SEN-04a early; twisted and shielded leads; Mode 0 fallback	T-SEN-04a fails at week -4: ship Mode 0 and the AI loop only
P-03	Safety tests reveal a hardware flaw (trip level, latch, reset state)	L	H	Run T-HW-03 to 10 first; keep two weeks for rework; keep a spare board	Any T-HW must test failing: stop all other work on it
P-04	Component delays	M	M	Order spares and the ADS1115 now; accept a local substitute that meets the same specification	Order not received by week -5
P-05	Cost above 575 TND	M	M	Weekly BOM tracker; mark optional items outside the cap	Running total above 520 TND
P-06	Volunteer recordings delayed by approvals	M	M	Request institutional approval and prepare a consent form now; keep all volunteer tests non-invasive and without stimulation	No approval at week -5: use team members as volunteers if the institution allows
P-07	Controller module mismatch (D-01)	L	H	Resolve D-01 this week	Still open at week -6
P-08	Scope creep (tuning, force servo, hosted database)	H	M	B.2 won't-have list; change control in B.3	Any request not in B.2
P-09	Team availability and exams	M	M	One owner and one backup per workstream; weekly checkpoint	Two checkpoints missed
P-10	Demonstration environment fails	M	M	No dependency on venue network; pre-recorded video; spare laptop and cables; printed evidence pack	Rehearsal fails twice
P-11	Audience reads more into the claims than the evidence supports	M	H	Claims register (B.8); scripted wording; limitations slide	Any slide with words from the never-say list
P-12	Battery or overheating event during testing	L	H	Protected cell, supervised charging, thermal test, fire-safe bag	Case above 45 °C: stop and inspect
P-13	Firmware regression after late changes	M	H	Configuration freeze and regression matrix (B.3), version tag on every build	Any change after the freeze date
P-14	Loss of code or recordings	L	M	Remote repository, second copy of the evidence folder, naming convention (B.8)	No backup in the last 7 days

L is likelihood and I is impact (L low, M medium, H high), the author's estimates.

Cut-line: what to drop first

1. Could-haves (Arabic strings, thigh IMU, load cell mount).
2. Should-have tests that need volunteers (T-SEN-06, 07, T-AI-06).
3. Mode 1 in the demonstration: show Mode 0 and the fixed-dose fallback only, and say Mode 1 is built but unproven. This is the first big cut and the one forced by a failing T-SEN-04a.
4. BLE: use the USB-serial path.
5. Soak, usability and runtime tests, in that order.
6. **Never cut:** T-HW-03 to 10, T-FW-01 to 05, T-SYS-03 fault injection, T-SYS-06 rehearsals and the claims register.

B.10 Decision log

Twelve decisions are open. Each has a recommendation; the team writes the decision, the date and the owner in the last column and keeps the log in the baseline folder.

Decision	Issue	Recommended	Owner, by when	Decided
D-01 Controller module	The Phase 2 table says ESP32-S2 with two cores and BLE. The S2 has one core and no Bluetooth, so the two-task design with BLE needs a classic ESP32 or an ESP32-S3.	Write the exact module that is flashed (classic ESP32 or ESP32-S3) into the spec and the BOM. If the board really is an S2, BLE and the two-core split cannot work as written. <i>Alternative:</i> None that keeps the current firmware.	Firmware lead Now (blocks BOM and spec)	[] accepted [] changed Date: _____
D-02 Ceiling cap and current margins	The ceiling range is 10–60 mA, but the NMES module tops out at 57.6 mA and the firmware cut is 60 mA, so the top of the range has no margin.	Cap the approved ceiling at 50 mA and keep the chain $50 \times 1.10 = 55$ mA, module maximum 57.6 mA, firmware cut 60 mA, hardware trip 65 mA. <i>Alternative:</i> Keep 60 mA and raise the firmware cut and the hardware trip, which means re-validating the trip.	Team and clinical adviser Before design freeze	[] accepted [] changed Date: _____
D-03 Reference MVC	The Phase 2 spec normalises EMG to an MVC recorded at the start of every session. Early after surgery, pain and inhibition make that value low and unrepeatable.	Record the reference MVC at the clinic (best of three efforts). Update it upward only, and only when the clinician records a higher value. Home sessions use the stored reference. <i>Alternative:</i> Per-session MVC: faster, but unreliable and unsafe when pain limits the effort.	Clinical adviser Before T-SEN-03	[] accepted [] changed Date: _____
D-04 Force sensing	EMG is not force. Mode 2, extremum-seeking tuning and any statement about force need a force or torque signal.	Leave a mounting point on the shank shell for a load cell with an HX711, but do not fit one in the MVP. <i>Alternative:</i> Fit a load cell now for validation only, which adds cost and a calibration task.	Team Design freeze	[] accepted [] changed Date: _____
D-05 Second IMU on the thigh	One MPU6050 on the shank gives shank motion. It cannot give hip flexion or trunk lean, and it has no magnetometer.	No second IMU in the MVP. Use the seated-posture instruction, the potentiometer and the shank-still flag, and add a thigh IMU if T-SEN-05 shows posture errors. <i>Alternative:</i> Add a second MPU6050 on the thigh cuff (address 0x69 with AD0 high).	Team Before T-SEN-05	[] accepted [] changed Date: _____
D-06 Plan approval cadence and expiry	Approval before each of the two daily sessions is heavy for clinicians. Reusing the last plan with no expiry is unsafe.	One approval per day covers both sessions. A plan is valid for 36 h by default (clinician range 12–168 h). The second session never exceeds the first one's ceiling. Pain of 4/10 or more, or a pain-related STOP, blocks any increase until review. <i>Alternative:</i> Per-session approval as in the Phase 2 spec, or auto-renewal rules after stable weeks (after the MVP).	Clinical adviser Before dashboard freeze	[] accepted [] changed Date: _____
D-07 ON time, ramps and duty cycle	Tables say ramps of 3 s. The state machine says 3 s up, 15 s hold, 2 s down. "ON 15 s" could mean the hold or the total. The protocol duty is about 1:3.	Freeze as built: 3 s up, 15 s hold, 2 s down (20 s of current). Duty is 1:1.5 at OFF 30 s, 1:2.25 at 45 s, 1:3 at 60 s and 1:4.5 at 90 s. <i>Alternative:</i> Hold of 10 s (15 s total), or a 3 s ramp down.	Team and clinical adviser Before T-HW-01	[] accepted [] changed Date: _____
D-08 Fallback to BLE	The BLE link is unfinished and the demonstration depends on the end-to-end flow.	Add a USB-serial path in the app or a PC tool for plan download and session upload, using the same plan format and integrity checks. It works between sessions only, because USB present means no stimulation. <i>Alternative:</i> Finish BLE only and accept the schedule risk.	Firmware and app leads Week -4	[] accepted [] changed Date: _____
D-09 Backend hosting for the demonstration	A hosted database needs security work that is not planned for the MVP.	Run the backend locally for the MVP, with anonymised patient IDs. Host it after the MVP. <i>Alternative:</i> Host now, which brings IR-06 and T-SW-06 into the must list.	Software lead Design freeze	[] accepted [] changed Date: _____

Decision	Issue	Recommended	Owner, by when	Decided
D-10 Patient-facing languages	Patients in Tunisia will use Arabic or French. The demonstration audience may use English.	Keep all patient strings in a table now. Show English and French in the demonstration. Add Arabic with a right-to-left layout after the MVP. <i>Alternative:</i> English only until the pilot.	App lead Could-have	☐ accepted ☐ changed Date: _____
D-11 Plan integrity	A plan received over BLE or USB could be corrupt, stale, for another patient or altered.	MVP: patient ID, increasing version number, expiry time and a CRC-32 checked by the brace. After the MVP: an HMAC with a per-brace key provisioned at the clinic. <i>Alternative:</i> CRC only: detects corruption but not tampering.	Firmware and backend leads Before T-FW-05	☐ accepted ☐ changed Date: _____
D-12 Usage-limit enforcement	"Maximum 2 sessions per day" is a user instruction in the checklist, not a firmware rule.	The brace refuses a third session in a day and a second session less than 3 h after the first, using a stored counter and a sanity-checked clock. An invalid clock refuses the session and asks for a clinic reset. <i>Alternative:</i> Advisory reminder in the app only.	Team and clinician Before T-FW-06	☐ accepted ☐ changed Date: _____

APPENDICES

Appendix A. Traceability matrix

Every requirement, grouped by subsystem, with its specification section, the tests that verify it and its evidence status. A requirement becomes **V Verified** only when its tests have passed against the stated criterion and the records are in the evidence pack. IDs MF and CF continue the numbering of v2; MF8 to MF10 and CF9, CF10 are new. Tests are defined in B.5.

Sensing (5)

ID	Requirement	Section	Tests	Status	Note
MF1	Measure quadriceps activation: RMS envelope of the surface EMG in % of the reference MVC, accepted only when the gate is open	V.3	T-SEN-01, 03, 04a	I Implemented	EMG module tested on a real muscle (clean RMS and median frequency); the ± 5 % MVC tolerance is not yet measured
MF2	Measure knee angle over 0–120° at 100 Hz, within $\pm 3^\circ$ of a goniometer, and count repetitions	IV.5, V.1	T-SEN-02, 07	I Implemented	Filter tuned against a goniometer; ESP32 ADC is non-linear; ADS1115 planned
MF3	Assess fatigue and movement quality: median-frequency drop in valid windows, IMU jerk, Model A state per contraction (≥ 90 % correct)	V.1, VIII.1	T-SEN-06, T-AI-01	S Simulated	93.5 % on simulated data
MF8	New: gate every measurement and every contraction start (knee $60^\circ \pm 10^\circ$, shank still, stimulator off, EMG valid)	V.3	T-SEN-05	D Planned	New in v3.0
FW-06	ATTEMPT and POSTURE_CHECK states before every contraction	VI.3, VII.1	T-SEN-05, T-SYS-02	D Planned	New states

Control (2)

ID	Requirement	Section	Tests	Status	Note
MF4	Deliver controlled NMES: plateau current set once per contraction between floor and ceiling, supervised every 100 ms, delivered current within ± 10 % of command	VI.1–VI.3	T-HW-02, T-FW-07, T-SYS-02	I Implemented	Firmware complete; closed-loop behaviour not yet bench-verified
FW-05	Assist rule with floor, ceiling, rate limit and fallback to Mode 0	VI.2	T-FW-07, T-SYS-02	D Planned	Rule is new; current firmware computes a PID every 100 ms

Stimulation and safety (13)

ID	Requirement	Section	Tests	Status	Note
CF1	NMES parameters locked in firmware: 50 Hz, 150–400 μ s, ramps 3 s up and 2 s down, hold 15 s, OFF 30–90 s, ceiling ≤ 50 mA	VI.4	T-HW-01, T-FW-01	I Implemented	Firmware refuses other values; waveform fidelity not yet recorded
CF2	Electrical safety: battery-only stimulation, isolated control lines, current limits, lead-off detection	X.1	T-HW-04 to 11	I Implemented	Implemented; fault injection on the assembled device not yet run
CF3	Emergency stop: hardware cut in less than 10 ms, latched until restart	X.1	T-HW-03	I Implemented	Latency not yet measured on the assembled brace
SR-01	Parameter limits enforced in firmware: any out-of-range value refused, last valid plan kept	X.1	T-FW-01	I Implemented	
SR-02	Approved ceiling (≤ 50 mA cap) clamps every command	VI.4, X.1	T-FW-07, T-HW-02	I Implemented	Cap is decision D-02
SR-04	Current tracking: error above 20 % or 5 mA for 3 pulses stops the burst	X.1	T-FW-04, T-HW-02	I Implemented	
SR-05	Over-current: firmware cut above 60 mA, hardware comparator above 65 mA, enable cut in less than 1 ms	IV.5, X.1	T-HW-04, 05	I Implemented	

ID	Requirement	Section	Tests	Status	Note
SR-06	Lead-off and open-circuit detection stops stimulation	X.1	T-HW-08	I Implemented	
SR-07	USB lockout: charge-only mode, no stimulation while USB is present	X.1	T-HW-06	I Implemented	
SR-08	Hardware watchdog: no heartbeat for 100 ms cuts the enable line	X.1	T-HW-07	I Implemented	
SR-09	Battery below 3.3 V ends the session; output is off at power-up, reset and brown-out	X.1	T-HW-09, 10	I Implemented	Reset state not yet measured
SR-10	Control lines galvanically isolated from the stimulation domain	VI.5	T-HW-11	I Implemented	
SR-13	Skin-contact surfaces at or below 43 °C in the worst-case session	X.1	T-HW-12	D Planned	New in v3.0

Firmware (6)

ID	Requirement	Section	Tests	Status	Note
MF10	New: enforce usage limits (at most 2 sessions a day, minimum gap, plan expiry)	VII.3	T-FW-05, 06	D Planned	New in v3.0
SR-11	New: stale-data and sensor-fault detection (EMG, angle or IMU missing or frozen for more than 500 ms)	X.1	T-SYS-03	D Planned	New in v3.0
FW-01	Session state machine with SAFE_STOP on any fault; restart only by explicit reset	VII.1	T-FW-02	I Implemented	
FW-02	Real-time and service tasks on separate cores; control period 100 ms ± 5 ms	VII.1	T-FW-03	I Implemented	Jitter under BLE load not yet logged
FW-03	Plan store: patient ID, version, expiry and checksum checked before use	IX.2	T-FW-05	D Planned	New in v3.0
FW-04	Session logging with time stamps and firmware version	VII.1	T-FW-08	I Implemented	

Software and data (10)

ID	Requirement	Section	Tests	Status	Note
MF6	Let the clinician validate every parameter; 100 % of sessions use an approved plan	IX.2	T-SW-02, 03	I Implemented	Dashboard and backend run locally
MF7	Record and share sessions: summary to the app over BLE, comfort 0–3 and pain 0–10; no session lost after sync	IX.2	T-SW-01, 04	G Gap	BLE link not finished
CF4	Human oversight: every plan approved by a clinician; the last approved plan is kept within its expiry	IX.2	T-SW-02, T-FW-05	I Implemented	Expiry rule is new
CF9	New: data protection (role-based access, hashed passwords, expiring tokens, no personal data in logs)	IX.3	T-SW-06	G Gap	Local file and basic security only
IR-01	Patient app: approved plan, comfort and pain entry, progress, STOP guidance	IX.1	T-SW-04	I Implemented	
IR-02	Clinician dashboard: accept or override each value, reasons, decision log, charts	IX.1	T-SW-02, 07	I Implemented	
IR-03	Backend runs Model B, stores decisions, serves the approved plan	IX.2	T-SW-04	I Implemented	Local file store
IR-04	Brace to phone link over BLE, with a USB-serial fallback	IX.2	T-SW-01	G Gap	BLE unfinished; fallback is decision D-08
IR-05	Plan integrity and offline behaviour: tamper, replay and expiry rejected; last valid plan kept	IX.2	T-SW-03, 05	D Planned	New in v3.0
IR-06	Hosted database and secure transport	IX.3	T-SW-06	G Gap	Local for the demo (D-09)

AI layer (9)

ID	Requirement	Section	Tests	Status	Note
MF5	Suggest the next prescription (ceiling, OFF time, contractions) with reasons; ceiling within 5 mA in ≥ 90 % of cases	VIII.1	T-AI-02, 03	S Simulated	97 % in simulation
SR-03	AI suggestion rules: at most +10 mA per session, -5 mA after pain or more than 30 % guarding, clamp before display	VIII.2	T-AI-03	I Implemented	
SR-12	Fatigue and guarding labels from Model A are advisory; STOP and the current rules stay primary	VIII.1	T-AI-01	S Simulated	
AR-01	Model A decision tree on the ESP32 (≥ 90 % correct)	VIII.1	T-AI-01	S Simulated	93.5 % simulated
AR-02	Model B export gives the same outputs as the Python model	VIII.3	T-AI-02	V Verified	300 of 300 test cases identical
AR-03	Model B performance: ceiling within 5 mA 97 % (fixed protocol 56 %), OFF exact 86 %, contractions exact 93 %	VIII.1	T-AI-03	S Simulated	Simulated patients only
AR-04	Each suggestion shows its two strongest reasons from the tree paths	VIII.2	T-AI-04	I Implemented	
AR-05	Clinician decisions are logged and used to retrain Model B	VIII.2	T-AI-05	S Simulated	Acceptance 64 % to 92 % in simulation
AR-06	Model C flags patients likely to stay below 110° at week 6	VIII.1	T-AI-01	S Simulated	Error 2.2°, alert correct 90 % (simulated)

Mechanical and power (2)

ID	Requirement	Section	Tests	Status	Note
CF5	Comfort and fit for adults 55–85; patella and electrode sites visible; STOP reachable	IV.3	T-MECH-01, 02	I Implemented	Printed and assembled; fit not yet tested on volunteers
CF6	Autonomy: two sessions of up to 40 min per charge	IV.5	T-PWR-01	D Planned	Energy estimate only

System and clinical (9)

ID	Requirement	Section	Tests	Status	Note
MF9	New: demonstration mode (stimulation into a dummy load only, volunteer EMG drives the command)	B.7	T-SYS-02, 06	D Planned	New in v3.0
CF7	Cost at or below 575 TND per prototype	App. B	BOM tracker	G Gap	Prices not yet entered
CF8	Designed with reference to IEC 60601-1, IEC 60601-2-10 and ISO 14971; dummy-load tests before any human use	X.5	T-HW-01 to 13	D Planned	No conformity is claimed
CF10	New: a first-time user can fit the brace, start a session and find STOP without help	B.1	T-SYS-04	D Planned	New in v3.0
SY-01	End-to-end flow: brace, app, backend, approval, brace	IX.2	T-SW-04	G Gap	Blocked by BLE
SY-02	Fault-injection campaign on the assembled device	B.6	T-SYS-03	D Planned	Not yet run
SY-03	Healthy-volunteer EMG recordings and retraining on real clinician decisions	XI.3	T-SEN-01, T-AI-06	D Planned	
SY-04	Ethics approval and a supervised TKA pilot	XI.3	after the MVP	D Planned	
SY-05	Regulatory pathway: classification, standards, technical file	X.5	after the MVP	D Planned	To be confirmed with a regulatory adviser

Appendix B. Bill of materials and cost tracker

The cap is 575 TND per prototype (CF7). No prices are filled in: the team did not give them and none were looked up. Fill in unit prices from actual receipts. Items marked optional are outside the cap and shown separately.

- **Weekly check.** Update the total every week. At 520 TND the risk P-05 triggers: stop adding items and review.
- **Count everything that is bought,** including spares, test fixtures and the dummy-load parts, or state clearly that they are excluded and why.
- **Cost claim.** Say the tracker total. Do not quote a figure that is not in the tracker (B.8).

#	Item	Qty	Unit TND	Line TND	Source	Note
1	Controller: ESP32 module with two cores and BLE (D-01)	1				Exact module name to be written here
2	Surface-EMG module (ready-made)	1				Tested on a real muscle
3	Ag/AgCl electrodes for EMG	1 pack				Separate from stimulation electrodes
4	Stimulation electrodes, self-adhesive	2 packs				Size chosen for current density; spare pack
5	NMES/EMS module, constant current, biphasic, 3.6–57.6 mA	1				Commercial
6	Optocouplers for control lines	6				Includes spares
7	Enable AND gate IC	2				Includes a spare
8	Shunt resistor and comparator for the hardware trip	2 sets				Trip at 65 mA
9	Latching STOP button	1				Red, within one-handed reach
10	Hardware watchdog (supervisor IC or retriggerable timer)	2				100 ms
11	IMU MPU6050 (GY-521)	1				Shank shell
12	Rotary potentiometer, 10 kΩ, single turn, low torque	2				One spare
13	ADS1115 16-bit ADC module	2				For the potentiometer; one spare
14	Cell 18650, protected	2				One spare
15	TP4056 USB-C charger module with protection	2				Includes a spare
16	PETG and TPU filament	1 kg				Orthosis shells and padding
17	Textile straps, hinge hardware, screws	1 set				
18	PCB or perfboard, connectors, wire, heat-shrink	1 set				
19	Dummy-load resistors (500 Ω, 1 kΩ, 2 kΩ, 1 W or more, 250 V rating)	1 box				For every stimulation test and the demonstration
20	Test fixtures: angle jig, sense resistor	1 set				T-SEN-02, T-HW-02
21	Optional (Tier 2): load cell with HX711 (D-04)	1				Outside the cap if needed
22	Optional (Tier 2): second MPU6050 (D-05)	1				Outside the cap if needed
Total (excluding optional items)						Cap 575 TND

Appendix C. Parameter register

Every number used in the design, its value, why it has that value, and where it is enforced. “Design value” means chosen by the author and not taken from a datasheet or a protocol; each needs a test or a confirmation. This register is part of the configuration baseline (B.3).

Group	Parameter	Value	Basis	Enforced in
NMES	Frequency	50 Hz	Post-TKA protocol (50 Hz at home)	Firmware, fixed
NMES	Pulse width	250 μ s default, 150–400 μ s	Protocol 250–400 μ s; clinician sets once	Firmware range
NMES	Waveform	Biphasic, symmetric, constant current	Module specification	Module
NMES	Ramp up, hold, ramp down	3 s, 15 s, 2 s	As built (D-07)	Firmware, fixed
NMES	ATTEMPT window before each contraction	3 s, stimulator off	Design value (new)	Firmware
NMES	OFF time	30, 45, 60 or 90 s	AI suggests, clinician validates	Firmware range
NMES	Contractions per session	10, 15 or 20	AI suggests, clinician validates	Firmware range
NMES	Approved ceiling	10–50 mA	Cap per D-02; module maximum 57.6 mA	Firmware, backend, app
NMES	Floor fraction	0.6 of the ceiling (0.4–1.0)	Design value; 1.0 gives a fixed dose	Firmware, clinician
NMES	Maximum change between contractions	3 mA	Design value	Firmware
Schedule	Sessions per day, minimum gap	2, 3 h	Protocol; gap per D-12	Firmware
Schedule	Plan validity	36 h (12–168 h)	D-06	Firmware
Schedule	Duration	6 weeks from 48 h after surgery	Protocol	Clinician
Position	Seated, hip about 85°, knee 60°	Gate: knee 60° $\pm$ 10°	Protocol; gate is design value	Firmware gate
Sensing	Raw EMG	2 kHz, band-pass 20–450 Hz, notch 50 Hz	Spec	Firmware
Sensing	EMG envelope	RMS over 100 ms, hop 50 ms (20 Hz)	Corrected from the 10 Hz in v2	Firmware
Sensing	Median-frequency window	256 ms FFT	Spec	Firmware
Sensing	Potentiometer, IMU, fusion	100 Hz, α 0.95–0.98	Spec	Firmware
Sensing	Shank-still threshold	below 10 °/s for 1 s (to tune)	Design value, T-SEN-05	Firmware
Sensing	Guard time after stimulation	1.0 s before EMG is valid again	Design value; T-SEN-04b decides	Firmware
Sensing	Stale-data timeout	500 ms	Design value	Firmware
Control	Voluntary activation A_v	Highest 300 ms mean of the envelope in the ATTEMPT window, in % of reference MVC	Design value	Firmware
Control	Target activation A_target	30 % of reference MVC (clinician-set)	Design value	Plan
Control	Gains K_p, K_i, u_max, x_0	0.5, 0.1 per contraction, 0.5, 0.5	Design starting values, tuned on the bench	Firmware
Safety	Current tracking	Error above 20 % or 5 mA for 3 pulses	Spec	Firmware
Safety	Over-current	60 mA firmware, 65 mA hardware, cut in under 1 ms	Spec	Firmware and hardware
Safety	STOP	Under 10 ms, latched	Spec	Hardware
Safety	Watchdog	100 ms without heartbeat	Spec	Hardware
Safety	Battery	Below 3.3 V ends the session	Spec	Firmware
Safety	AI rules	+10 mA at most per session; –5 mA after pain of 4/10 or more or guarding above 30 %	Spec; pain threshold to confirm	Backend

Appendix D. Self-tuning study (simulation)

The extremum-seeking paper provided with the project (Roux-Oliveira et al., 2019) tunes the gains of a PID controller for NMES during use, by perturbing the gains and watching a cost. This appendix records what that idea can and cannot do in this device.

D.1 What the method is

$$\vartheta_i(k+1) = \vartheta_i(k) - \gamma a_i \sin(\omega_i k) \frac{J(k) - \bar{J}(k)}{\bar{J}(k)}, \quad \vartheta = [\ln K_p, \ln K_i]$$

Extremum-seeking update of the log-gains with perturbation amplitude a_i , frequency ω_i and gain γ ; J is the cost of the contraction and $\bar{J}$ its running mean.

$$J = \sum_{t \in \text{plateau}} e(t)^2 + \lambda_o (\text{overshoot})^2 + \lambda_I \bar{I} + \lambda_u (\text{chatter})$$

The cost used in the study: squared tracking error over the plateau, overshoot, mean current and chatter.

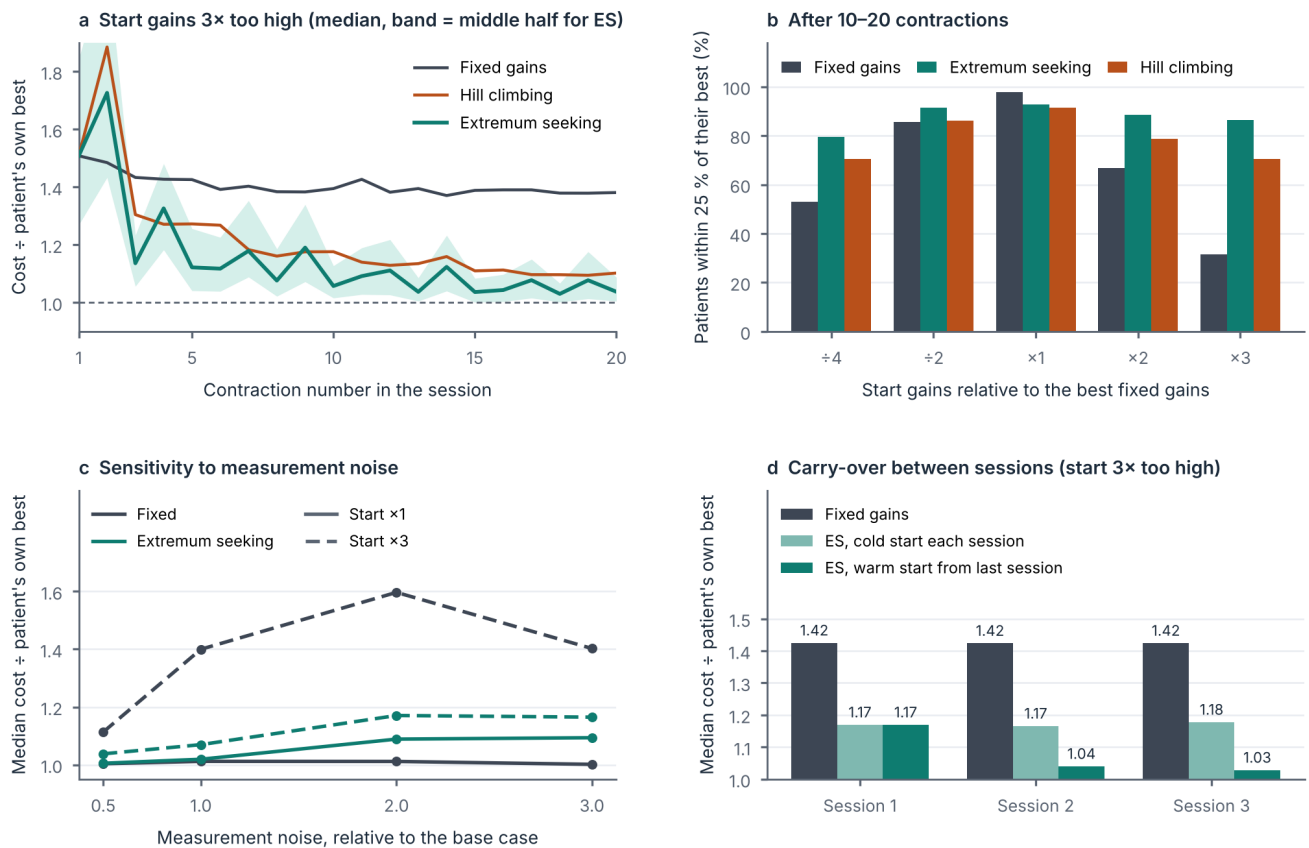
The paper reports eight stroke patients and no fixed-gain control group, so it does not show that the method beats well-chosen fixed gains.

D.2 Why it cannot tune the EMG assist rule

In Mode 1 the controlled signal is voluntary EMG read with the stimulator off. It does not depend on the current, so the cost J carries no information about the gains, and any perturbation would only be noise. The method needs a signal that the current actually changes: force or torque at the shank (a load cell), or the evoked EMG. That is why it belongs with Mode 2 (VI.1) and a force sensor (D-04). An earlier design note by the author said otherwise; that was wrong (Appendix E).

D.3 What a simulation shows

The study uses a toy plant, 400 test patients, and hyper-parameters tuned on 200 other patients. It compares fixed gains, hill climbing and extremum seeking over 20 contractions per session. It is not the team's simulator, not a model of a real muscle and not a clinical result; it shows the behaviour of the method on a plant built to resemble the controller's problem.



Toy plant, 400 test patients; hyper-parameters tuned on 200 other patients. Illustrative: not the team's simulator and not a clinical result.

Figure 10. Self-tuning study on a toy plant. (a) Convergence from gains 3× too high; (b) fraction of patients within 25 % of their best cost after 10–20 contractions; (c) sensitivity to noise; (d) carry-over between sessions.

- **Recovery from bad gains.** Starting with gains three times too high, 32 % of patients end within 25 % of their own best cost with fixed gains, 70 % with hill climbing and 86 % with extremum seeking. By contraction 16 the median cost has fallen from 1.51 to 1.04 times the patient's best with extremum seeking, against 1.39 for fixed gains.
- **No gain when the gains are already good.** With well-chosen starting gains, fixed gains put 98 % of patients within 25 % and extremum seeking 93 %: it can only match, and its perturbations cost a little.
- **Noise sensitivity.** At twice the base noise from well-chosen gains, extremum seeking's median cost is 1.09 against 1.01 for fixed gains. It helps most when the start is bad and hurts a little when it is good and the signal is noisy.
- **Carry-over helps.** Starting each session from the last session's gains gives a median cost of 1.04 and 1.03 in sessions 2 and 3, against 1.17 when it restarts cold and 1.42 with fixed gains.

NOTE · Conclusion

Extremum seeking is a recovery mechanism for badly set gains, not an improvement over well-set gains, and it needs a force or evoked-response signal. Recommended use, after the MVP: a supervised commissioning aid on a bench or at the clinic with a load cell, with warm starts across sessions and a clinician reviewing the gains it proposes. These are simulated results **S Simulated** and must not be quoted as device performance.

Appendix E. Change log from the Phase 2 specification

23 changes, each with the reason. The last two rows correct the author's own earlier design note.

Area	v2 (Phase 2)	v3.0	Why
Controller	ESP32-S2 listed as dual core with BLE	Exact module to be named; classic ESP32 or ESP32-S3 fits the design (D-01)	The S2 has one core and no Bluetooth

Area	v2 (Phase 2)	v3.0	Why
Ceiling	10–60 mA	10–50 mA cap with a margin chain (D-02)	Module maximum is 57.6 mA; the firmware cut is 60 mA
Timing	Ramps 3 s; ON 15 s	3 s up, 15 s hold, 2 s down, plus a 3 s ATTEMPT (D-07)	Tables and state machine disagreed
EMG envelope	10 Hz from 100 ms windows with 50 % overlap	20 Hz (100 ms windows, 50 ms hop)	Arithmetic: 50 % overlap gives a new value every 50 ms
EMG during stimulation	PID uses EMG every 100 ms while stimulating	EMG is used only when the stimulator is off; the 100 ms loop supervises current	Evoked response and artefact dominate EMG while current flows
Controller name	PID with derivative and anti-windup	PI-type assist rule updated once per contraction	The derivative has no use on a once-per-contraction EMG value
MVC	Recorded at the start of each session	Clinic-recorded reference, updated upward only (D-03)	Early post-operative effort is low and unrepeatable
Fatigue index	Median frequency over the whole session	Median frequency in valid windows only	Evoked EMG changes with fatigue differently from voluntary EMG
Lead-off rule	EMG flat for more than 1 s	Noise floor outside its calibrated window for more than 1 s	Early after surgery a healthy electrode can show almost no voluntary EMG
New states	—	POSTURE_CHECK and ATTEMPT before each contraction	Gates every measurement and start
New safety rules	—	Stale-data timeout, usage limiter, plan expiry and integrity, thermal limit	Gaps found in the audit
Plan approval	Before each of the 2 daily sessions; last plan kept with no expiry	One approval per day, 36 h expiry, no increase after pain (D-06)	Workload and stale-plan risk
Angle accuracy	$\pm 3^\circ$ with a $\pm 1\%$ potentiometer	Error budget with 5-point calibration and ADS1115	Linearity alone is about $\pm 2.7^\circ$ of a 270° travel
Standards	“Standards followed in the design”	“Designed with reference to”; related standards listed; no conformity claimed	Avoids over-claiming
Evidence	Mixed statements of what was built and tested	Every requirement carries V, I, S, D or G	Honest readiness for the jury
Section numbering	X, then XII	XI added: verification and roadmap	Numbering gap
Contraindications	List in the spec; shorter list in the checklist	One list, with precautions separated	Consistency
Usage limit	Instruction in the checklist	Firmware limiter (D-12)	An instruction is not a control
Backend rule	Uploads above 60 mA rejected	Above 50 mA rejected	Matches the cap
Dummy-load test range	10–60 mA	10–50 mA	Module maximum 57.6 mA
Form wording	“The AI could not slow down the stimulation”	“The AI must not add delay to the real-time loop”	Clarity
Self-tuning	Not in the Phase 2 spec	Roadmap: needs a force signal; simulation study in Appendix D	Voluntary EMG read while the stimulator is off cannot be tuned by the current
My earlier design note	Said the IMU sits on the thigh and that extremum seeking could tune gains from EMG read in gaps	IMU is on the shank; extremum seeking needs a force or torque signal	Both statements were wrong against the spec and the physics

Appendix F. Glossary

Term	Meaning
A_v	Voluntary activation index of one contraction: the highest 300 ms mean of the EMG envelope in the ATTEMPT window, in % of the reference MVC.
AMI	Arthrogenic muscle inhibition: reflex inhibition of a muscle by a painful, swollen or injured joint.
ATTEMPT	State before each contraction: stimulator off, the patient pushes into knee extension.

Term	Meaning
BLE	Bluetooth Low Energy.
DVT	Deep vein thrombosis.
EMG, sEMG	Electromyography; surface EMG records muscle electrical activity from the skin.
ES	Extremum seeking: an adaptive method that tunes parameters by perturbing them and watching a cost.
FMEA, RPN	Failure modes and effects analysis; risk priority number, severity × occurrence × detection difficulty.
Floor, ceiling	Lowest and highest plateau current allowed in a plan. The ceiling is approved by the clinician; the floor is a fraction of it.
ISE	Integral of squared error, the cost used in the extremum-seeking paper and in Appendix D.
MDF	Median frequency of the EMG power spectrum; it falls as a muscle tires.
MVC, MVIC	Maximal voluntary contraction; maximal voluntary isometric contraction.
NMES	Neuromuscular electrical stimulation.
PI	Proportional-integral controller.
RMS	Root mean square, the amplitude measure of the EMG envelope.
ROM	Range of motion of the knee.
SAFE_STOP	Latched firmware state that cuts stimulation in hardware until the brace is restarted.
TKA	Total knee arthroplasty (knee replacement).
TND	Tunisian dinar.

Appendix G. Sources and verification notes

This document was prepared with AI assistance from the team's Phase 2 specification and form answers. It does not carry out tests. The table says, for each source, what was checked and what was not.

Source	What was checked
Roux-Oliveira T, Costa LR, Pino AV, Paz P. Extremum Seeking-based Adaptive PID Control applied to Neuromuscular Electrical Stimulation. An Acad Bras Cienc 2019; 91(Suppl 1).	Provided PDF, read in full. Eight stroke patients; no fixed-gain control group.
Stevens-Lapsley JE, et al. Early neuromuscular electrical stimulation to improve quadriceps muscle strength after total knee arthroplasty: a randomized controlled trial. Phys Ther 2012; 92(2):210–226.	Cited in the team's Phase 2 specification; not re-read for this document.
University of Delaware. Unilateral total knee arthroplasty rehabilitation protocol (2021).	Cited in the team's specification; the page could not be fetched here, so protocol values are taken as stated by the team.
Mizner RL, Petterson SC, Stevens JE, Vandenborne K, Snyder-Mackler L. Early quadriceps strength loss after total knee arthroplasty. J Bone Joint Surg Am 2005; 87:1047–1053.	Abstract values checked: strength –62 %, voluntary activation –17 %, muscle cross-section –10 % at about one month.
Revisiting the application of neuromuscular electrical stimulation at the knee (review). PMC10872626.	Cited in the team's specification; not re-read.
Rehabilitation protocols for total knee arthroplasty: Mass General Brigham; Brigham and Women's Hospital; Lindsay guideline; Altru; Saudi Ministry of Health.	Used for the phase map only. Found by web search and read as excerpts; check the originals before relying on a number.
IEC 60601-1, IEC 60601-1-11, IEC 60601-1-2, IEC 60601-2-10, IEC 62304, IEC 62366-1, ISO 14971:2019, ISO 10993-1, IEC 62133-2, ISO 13485.	Titles from general knowledge; editions and clauses not checked. Confirm with the standards body and a regulatory adviser.
Component facts (ESP32 variants, ADS1115, MPU6050, HX711).	From general knowledge; the datasheet fetch was blocked. Check against the datasheets of the parts you hold.
Team, Technical Architecture Report (Phase 1), SymbioMed Challenge, IEEE TRSYN 3.0, 2026.	Cited in the Phase 2 specification; not provided here.

Not verified: the team must check these

- Datasheet values for the controller variants (S2 single core, no Bluetooth; classic ESP32 and S3 with two cores and Bluetooth), ADS1115, MPU6050 and HX711 are from general knowledge.
- Protocol values (Delaware, Stevens-Lapsley) are as stated in the team's specification; the source pages were not re-read here.
- Standard editions and clauses, and the EU and US classification of the device, are the author's reading and need a regulatory adviser.
- All pass criteria in B.5 and all design values in Appendix C are proposals, not tested numbers.
- FMEA ratings F-09 to F-19 and the project-risk likelihoods are the author's estimates.
- The simulation in Appendix D is a toy plant, not the team's simulator.

CAUTION · Not a medical device

This document describes a research prototype. It is not a medical device, no stimulation has been applied to a person, and nothing here is clinical advice or a claim of clinical benefit or safety for patients.